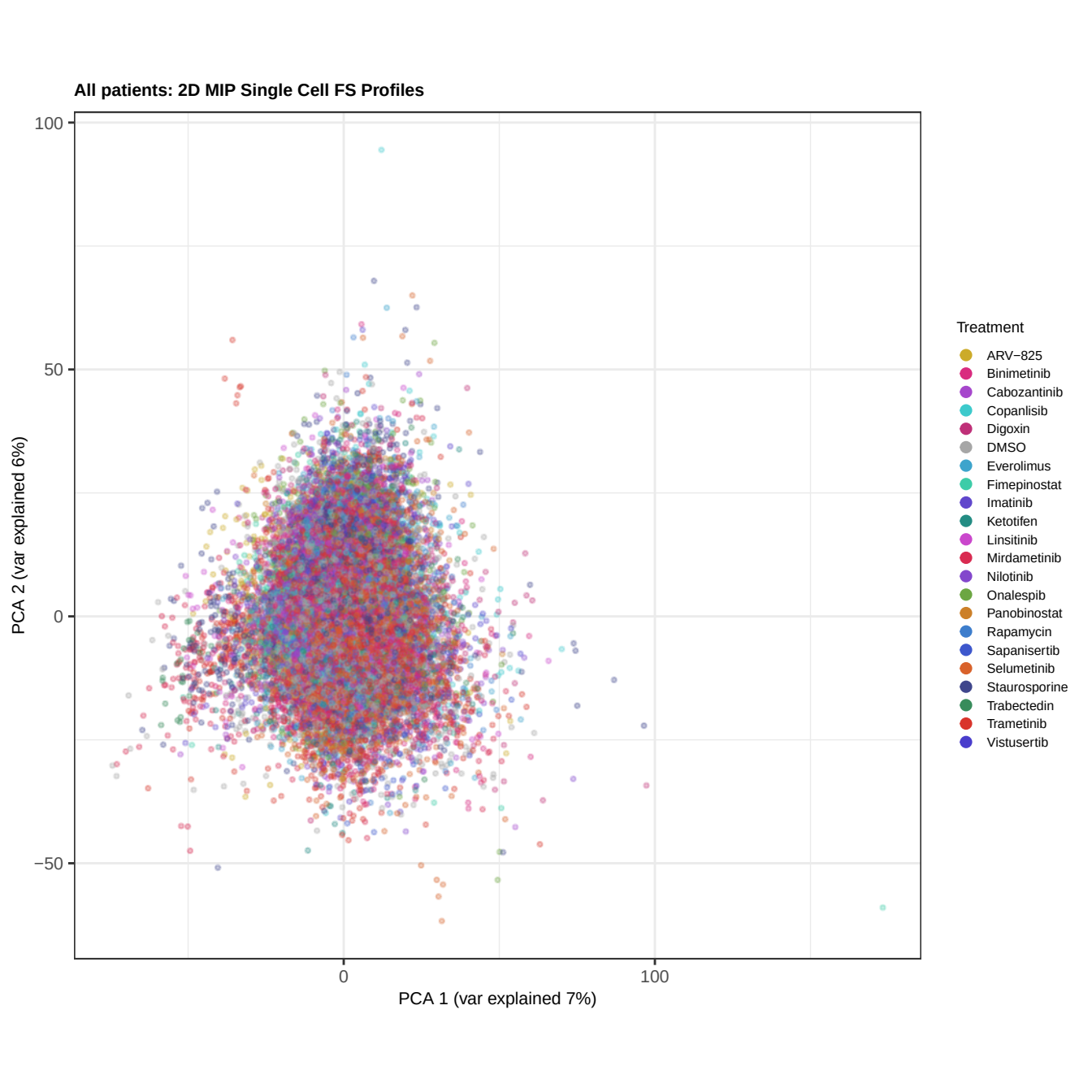
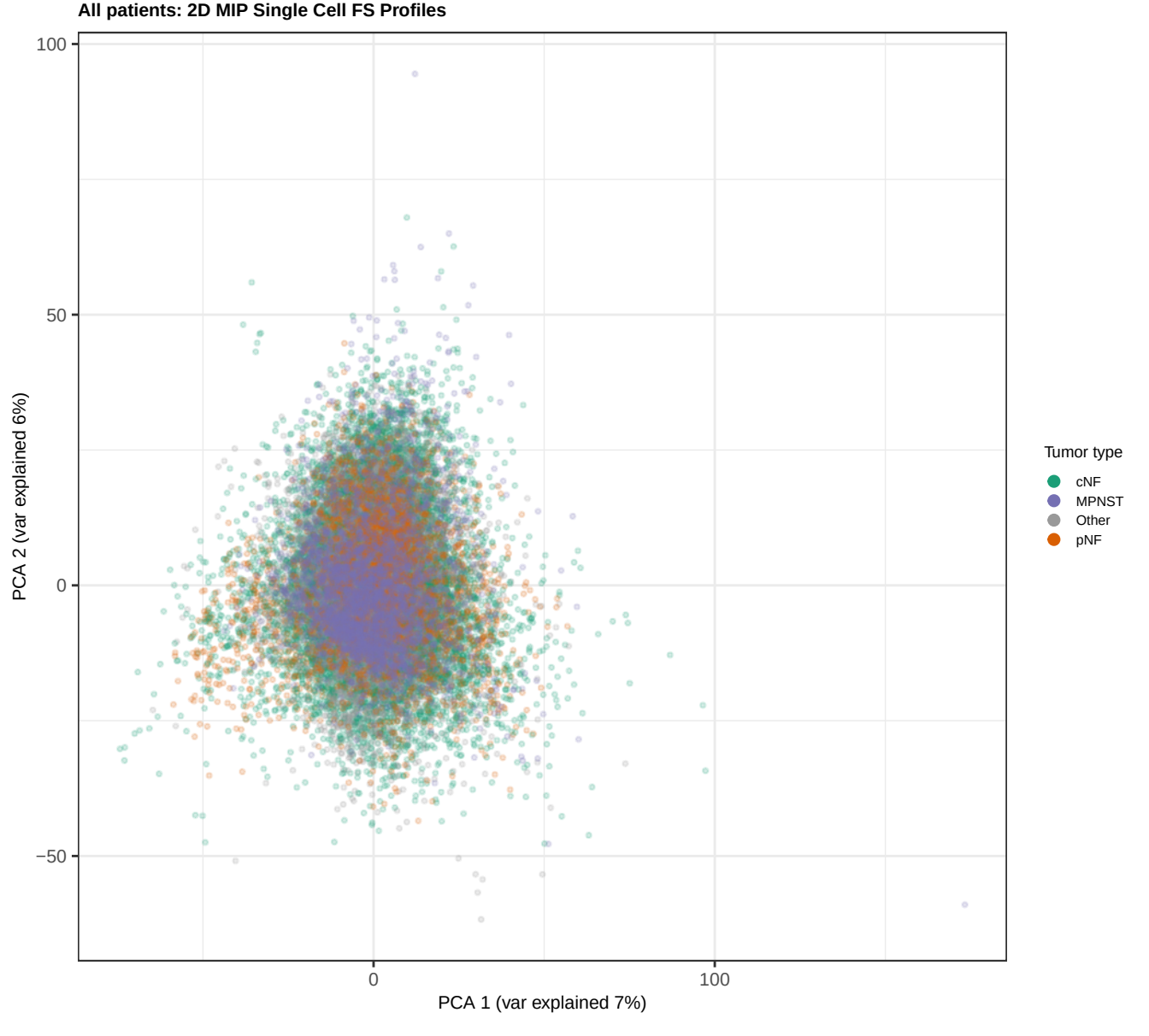
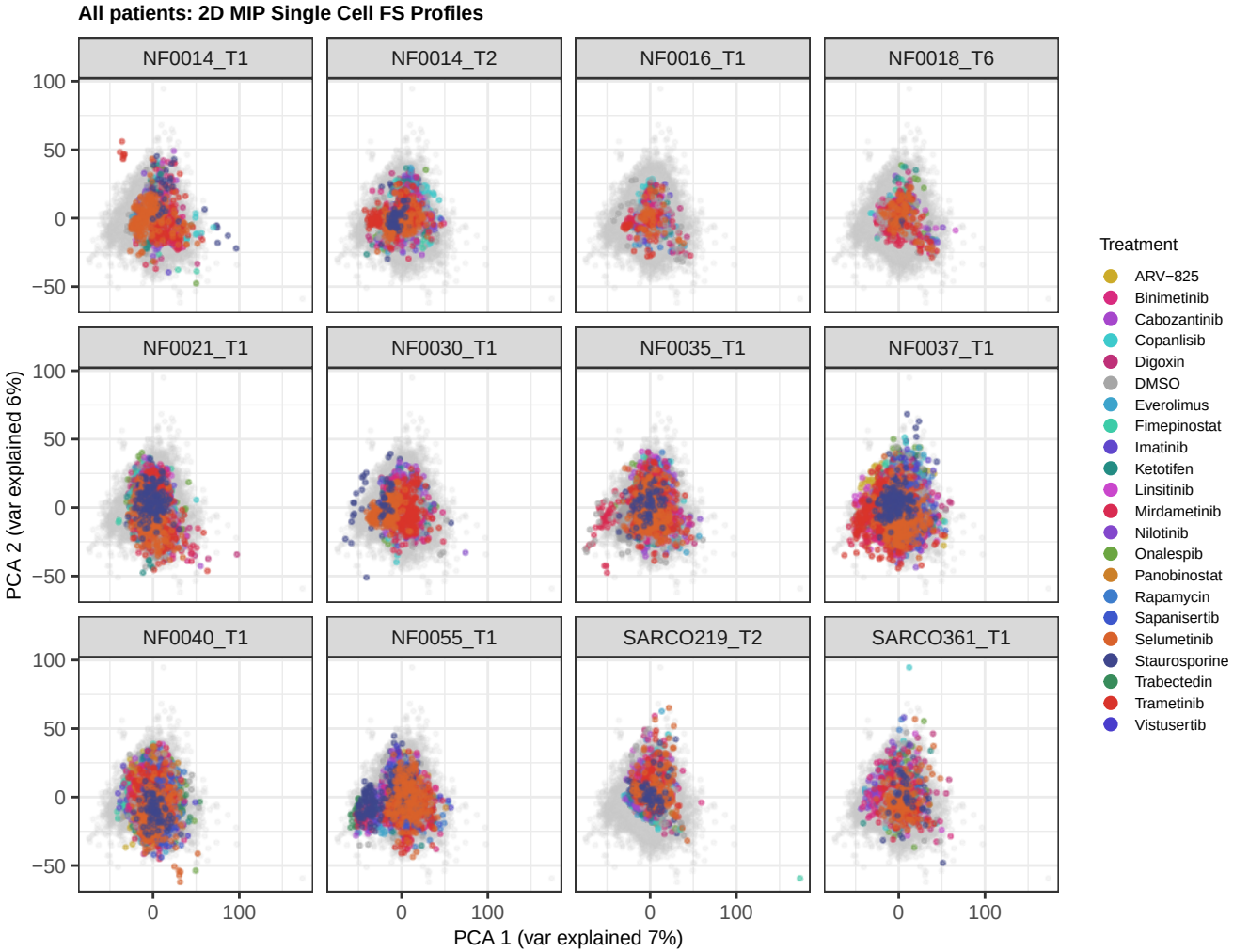
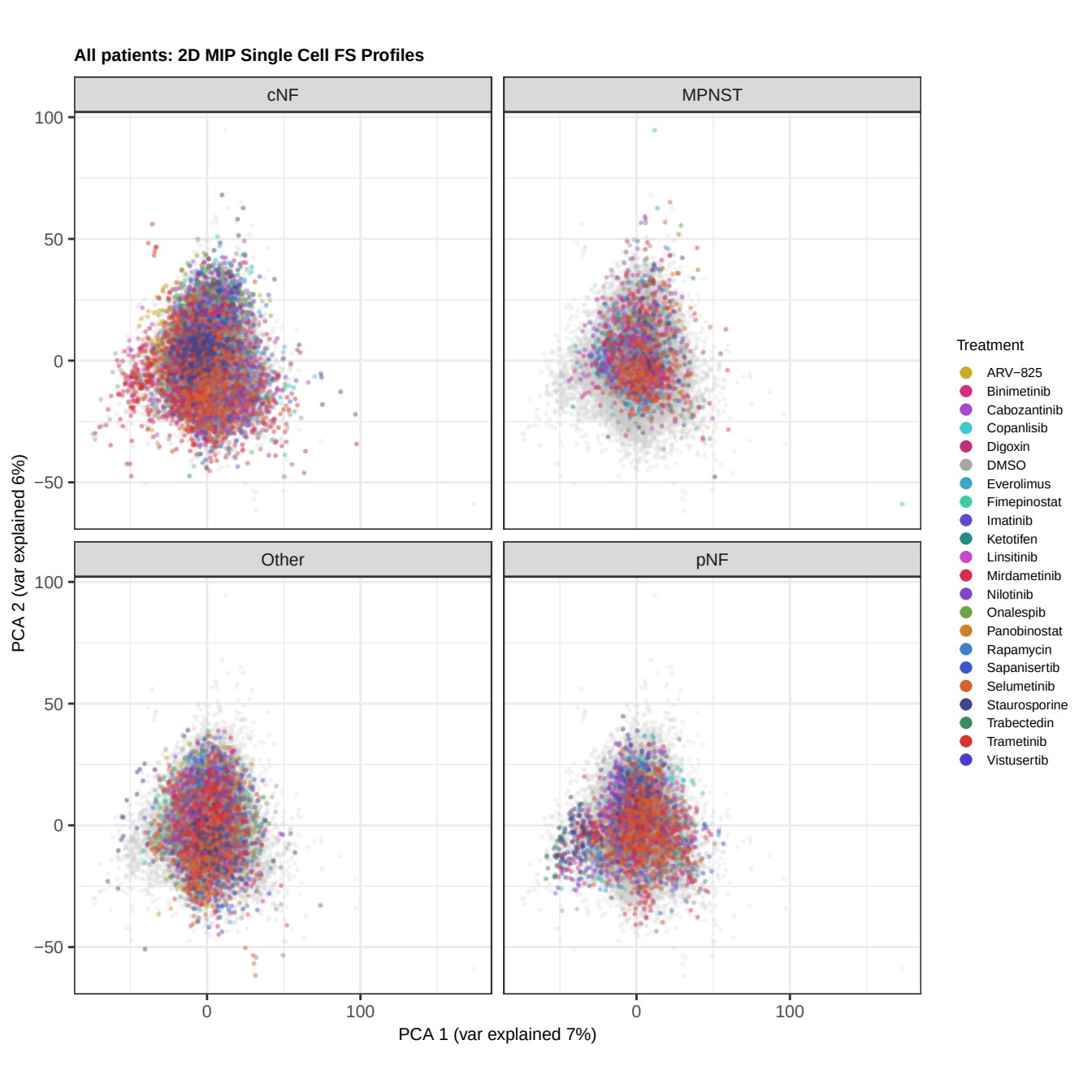
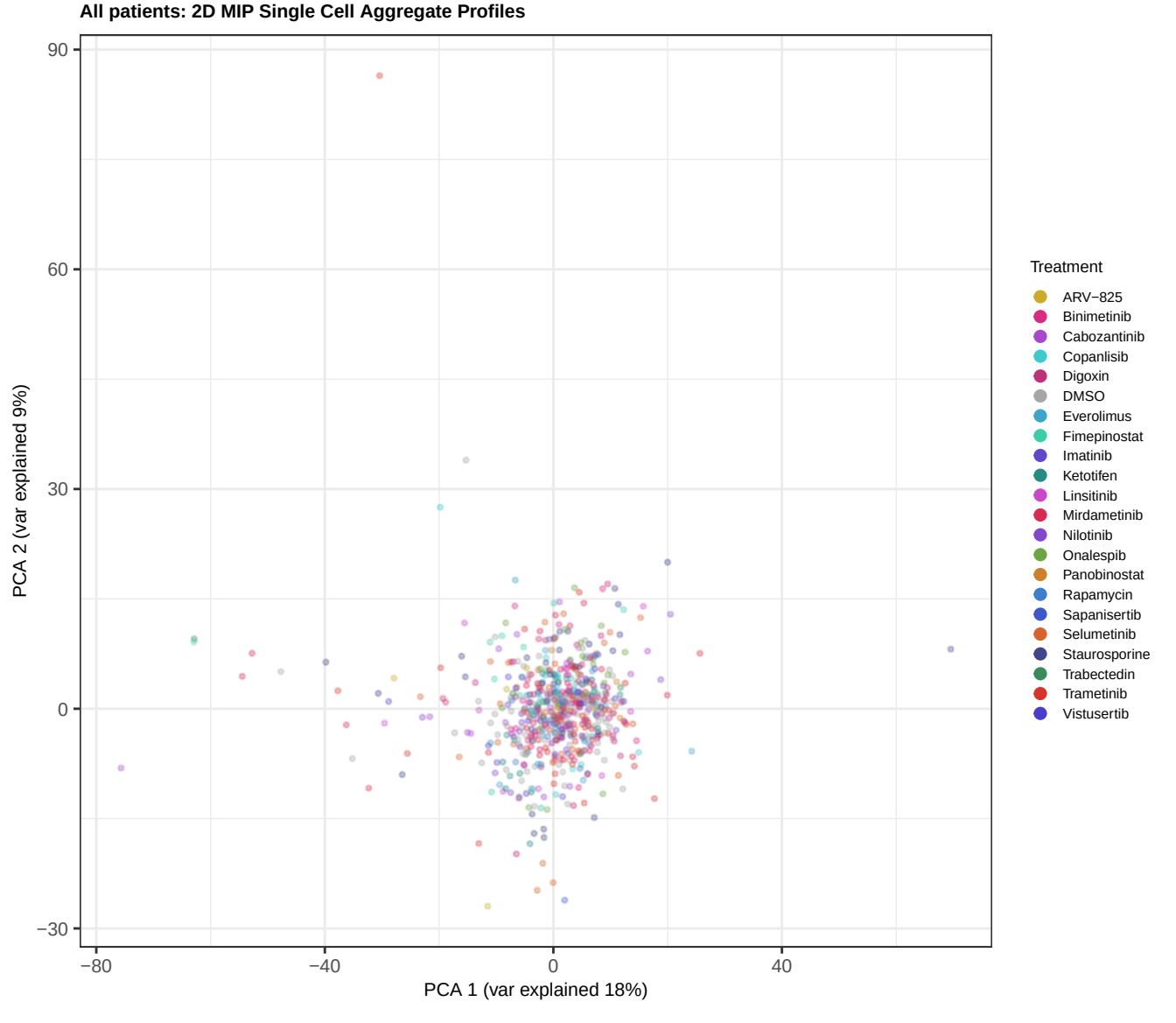
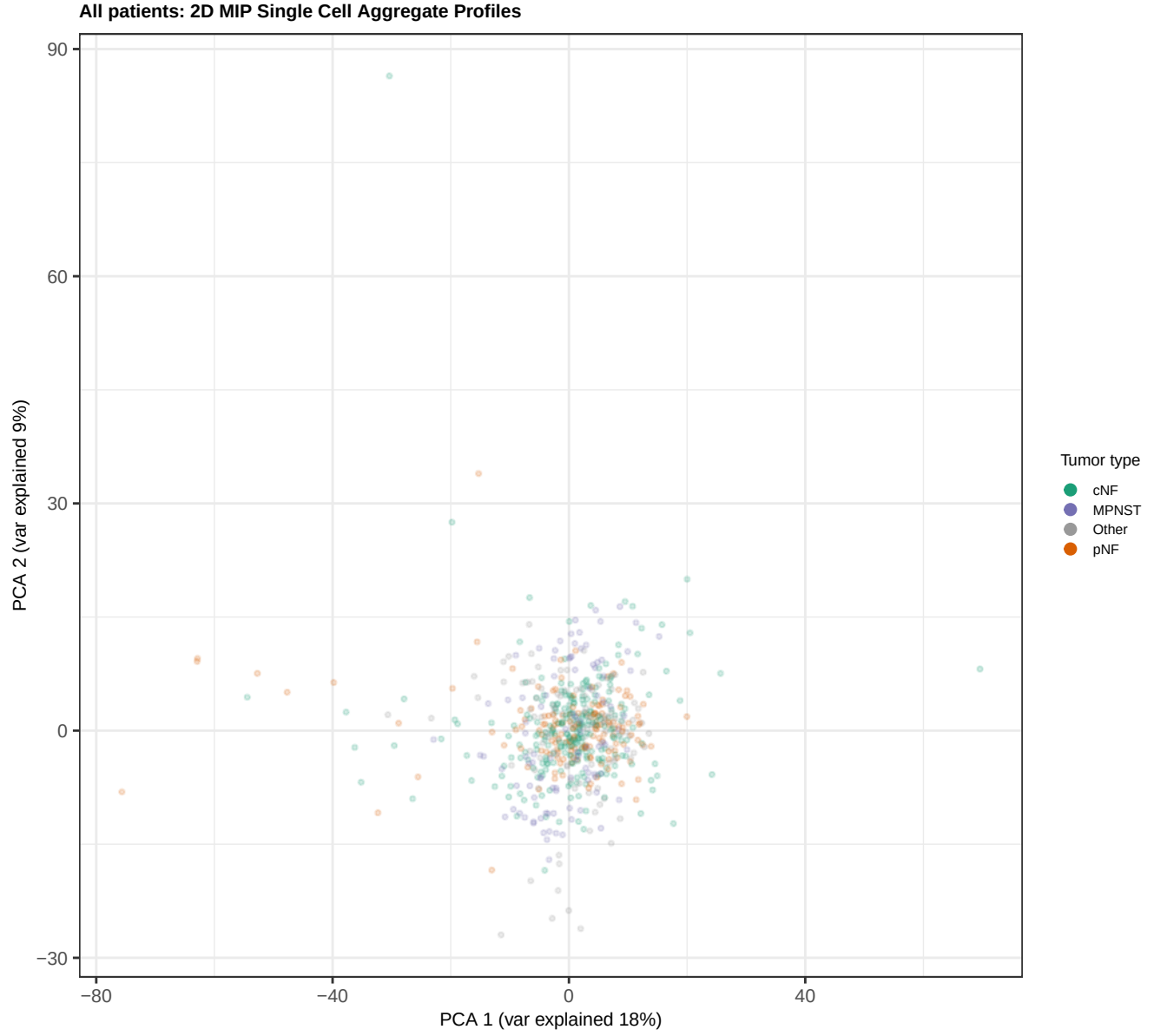

All patients: 2D MIP Single Cell FS Profiles
100
50
0
−50
PCA 2 (var explained 6%)
0
100
PCA 1 (var explained 7%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

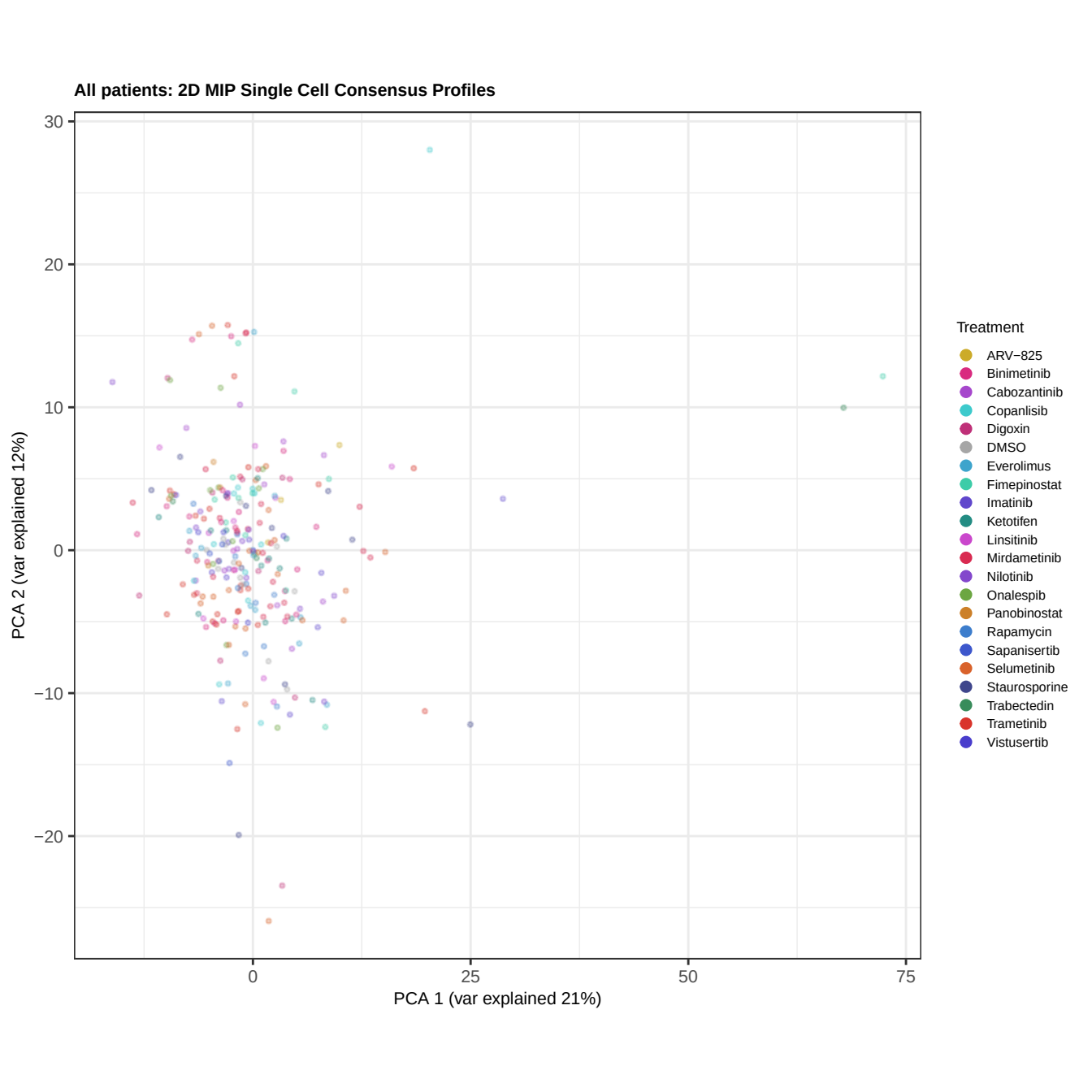

All patients: 2D MIP Single Cell FS Profiles
100
50
0
−50
PCA 2 (var explained 6%)
0
100
PCA 1 (var explained 7%)
Tumor type
cNF
MPNST
Other
pNF

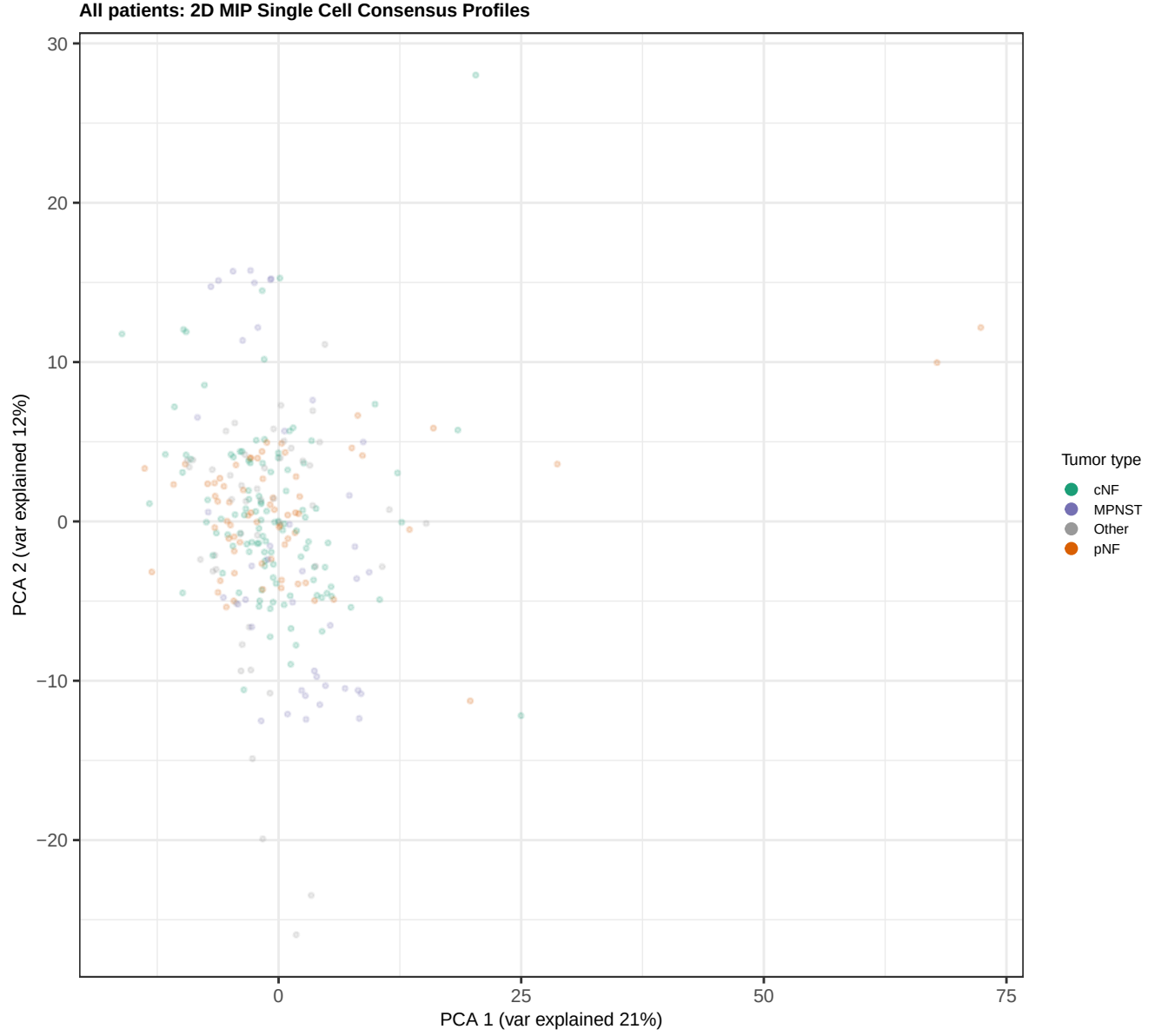

All patients: 2D MIP Single Cell FS Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
100
50
0
−50
0
100
PCA 2 (var explained 6%)
PCA 1 (var explained 7%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

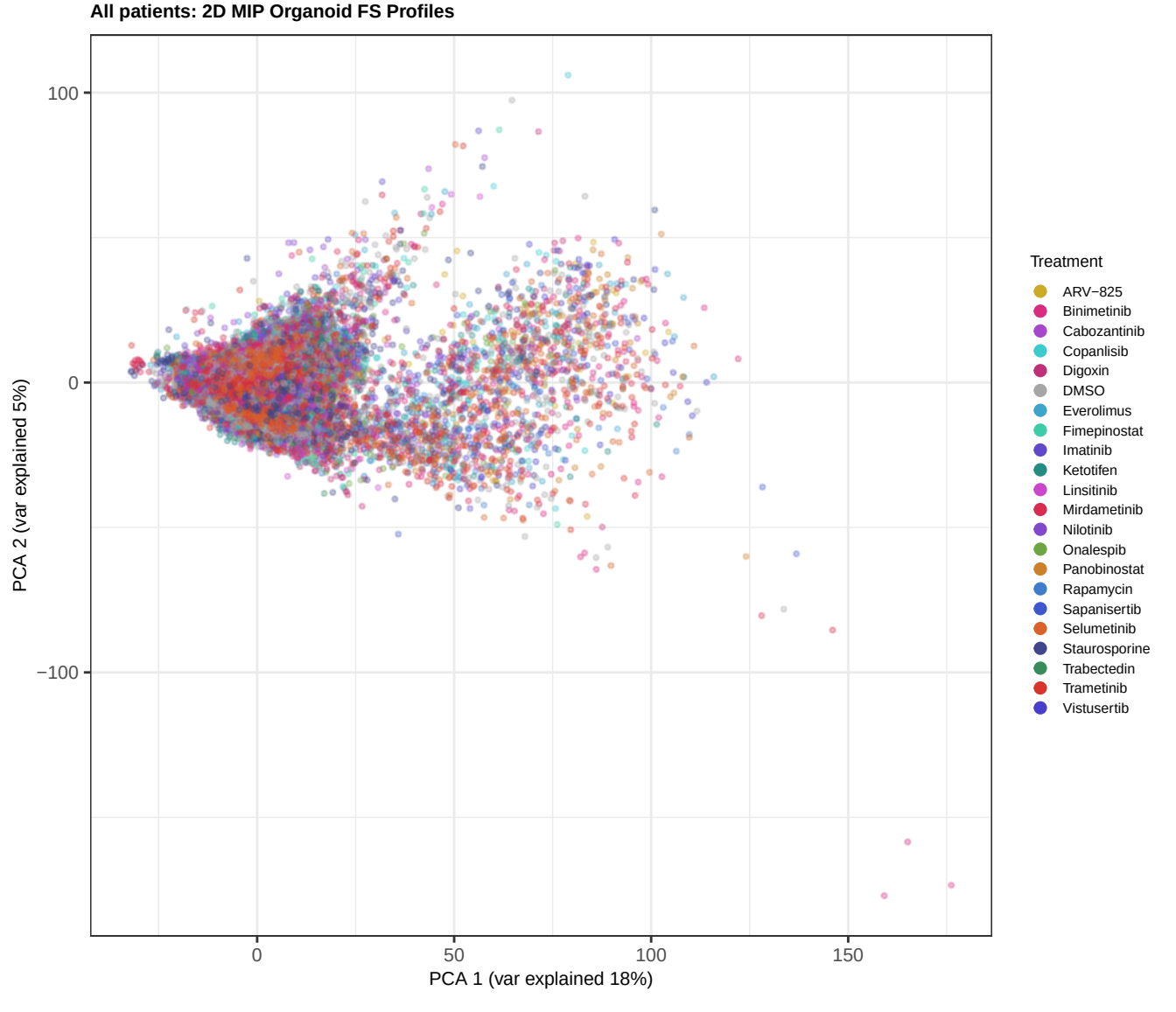

All patients: 2D MIP Single Cell FS Profiles
cNF
MPNST
Other
pNF
100
50
0
−50
PCA 2 (var explained 6%)
0
100
PCA 1 (var explained 7%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

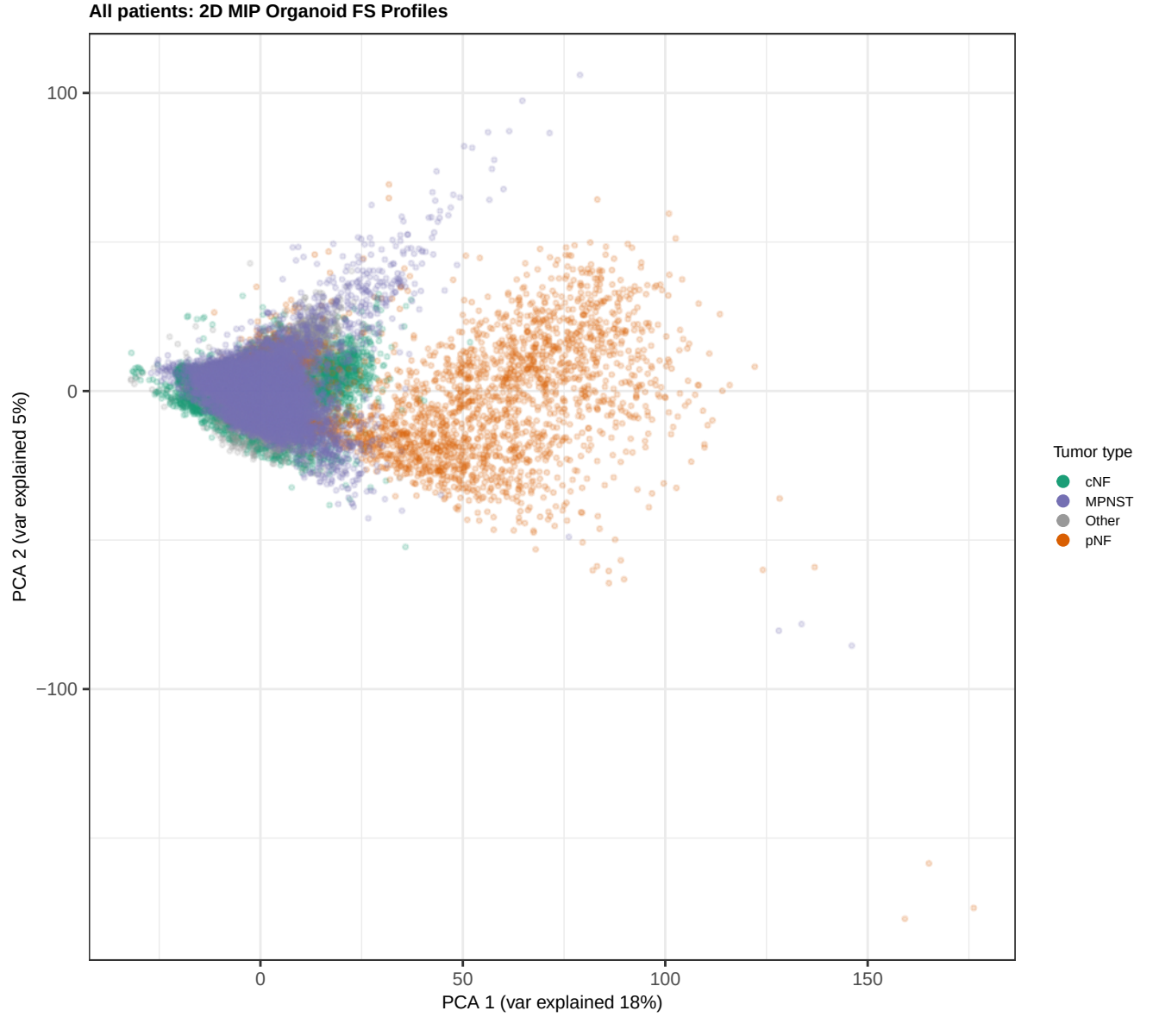

All patients: 2D MIP Single Cell Aggregate Profiles
90
60
30
0
−30
PCA 2 (var explained 9%)
−80
−40
0
40
PCA 1 (var explained 18%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

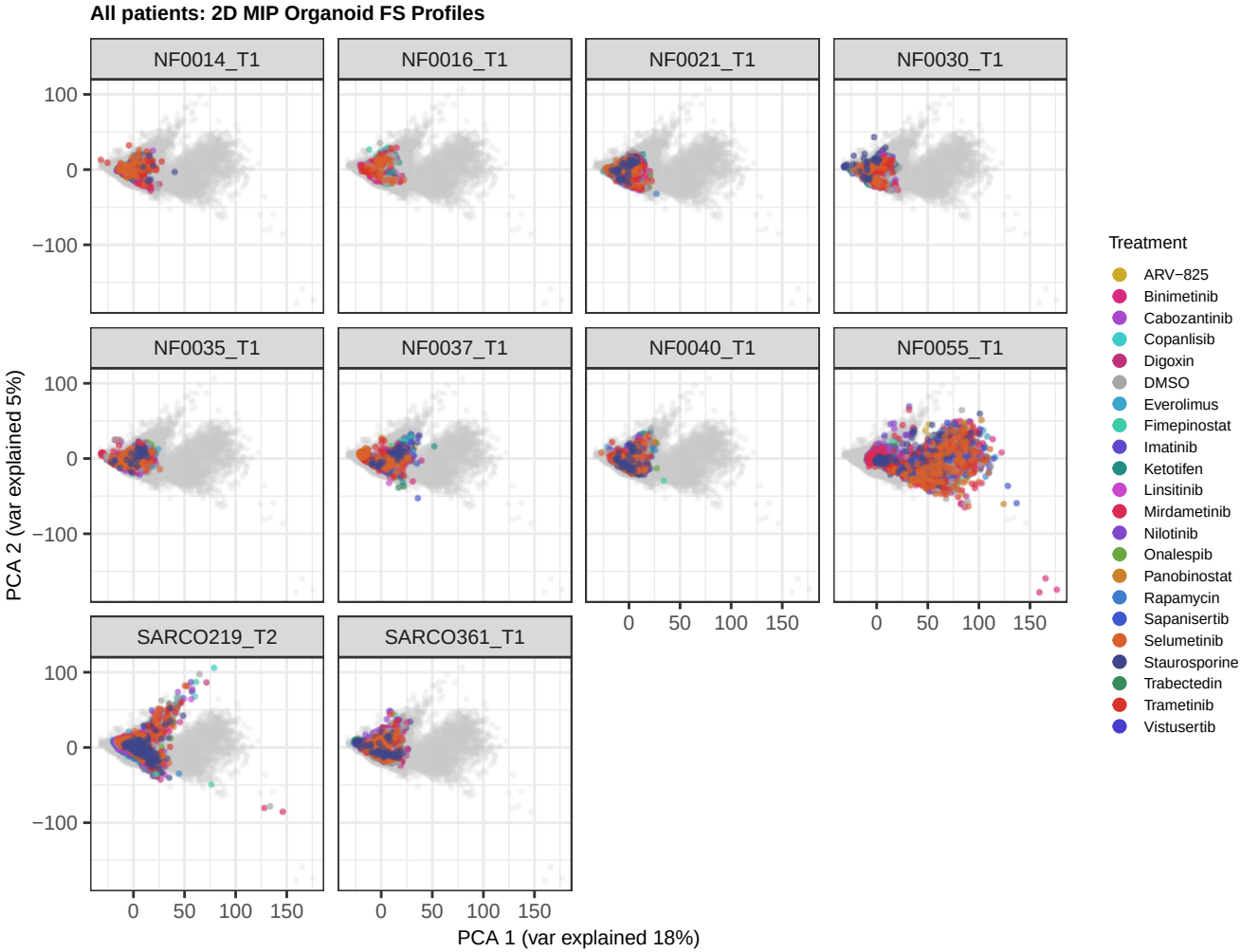

All patients: 2D MIP Single Cell Aggregate Profiles
90
60
30
0
−30
PCA 2 (var explained 9%)
−80
−40
0
40
PCA 1 (var explained 18%)
Tumor type
cNF
MPNST
Other
pNF

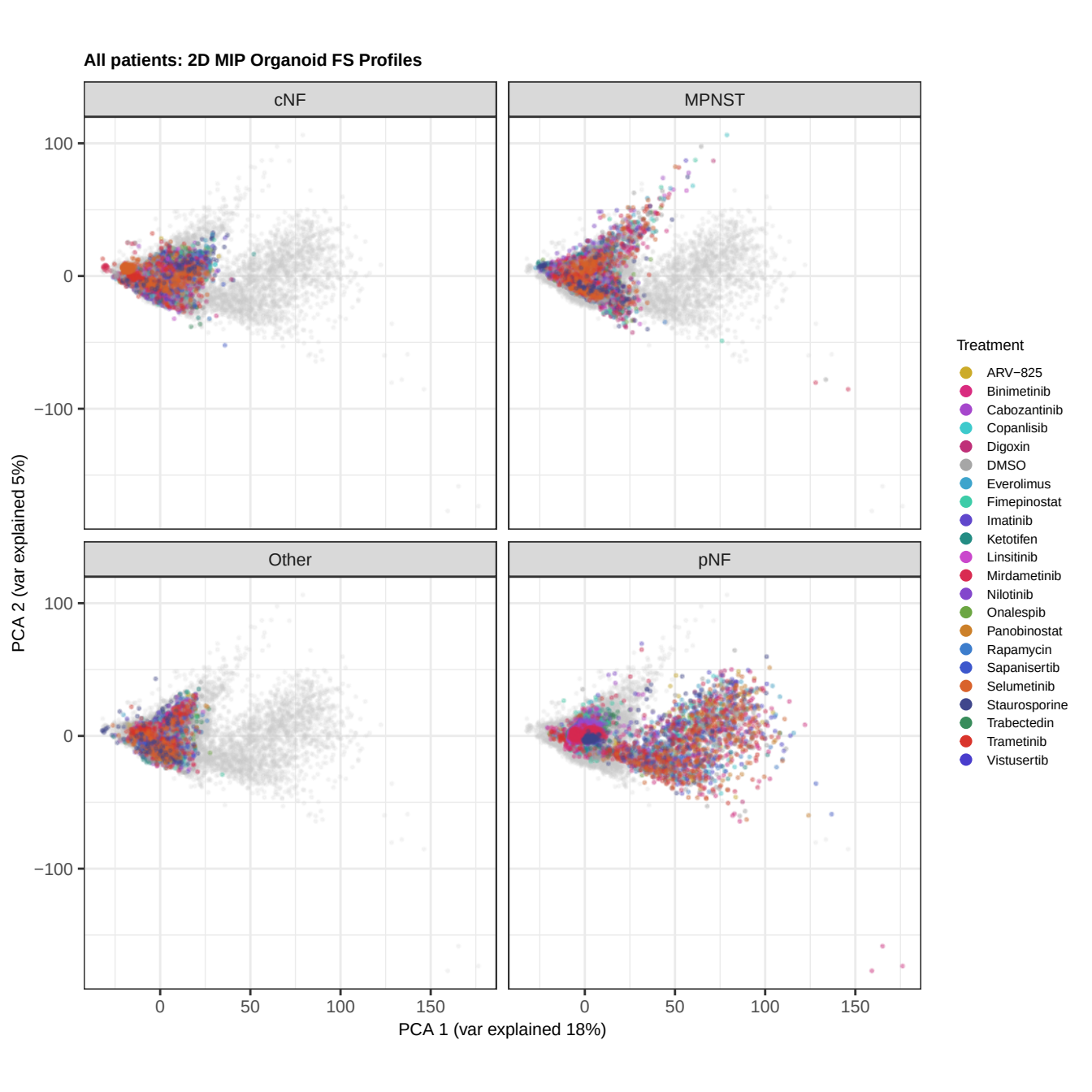

All patients: 2D MIP Single Cell Consensus Profiles
PCA 2 (var explained 12%)
PCA 1 (var explained 21%)
30
20
10
0
−10
−20
0
25
50
75
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

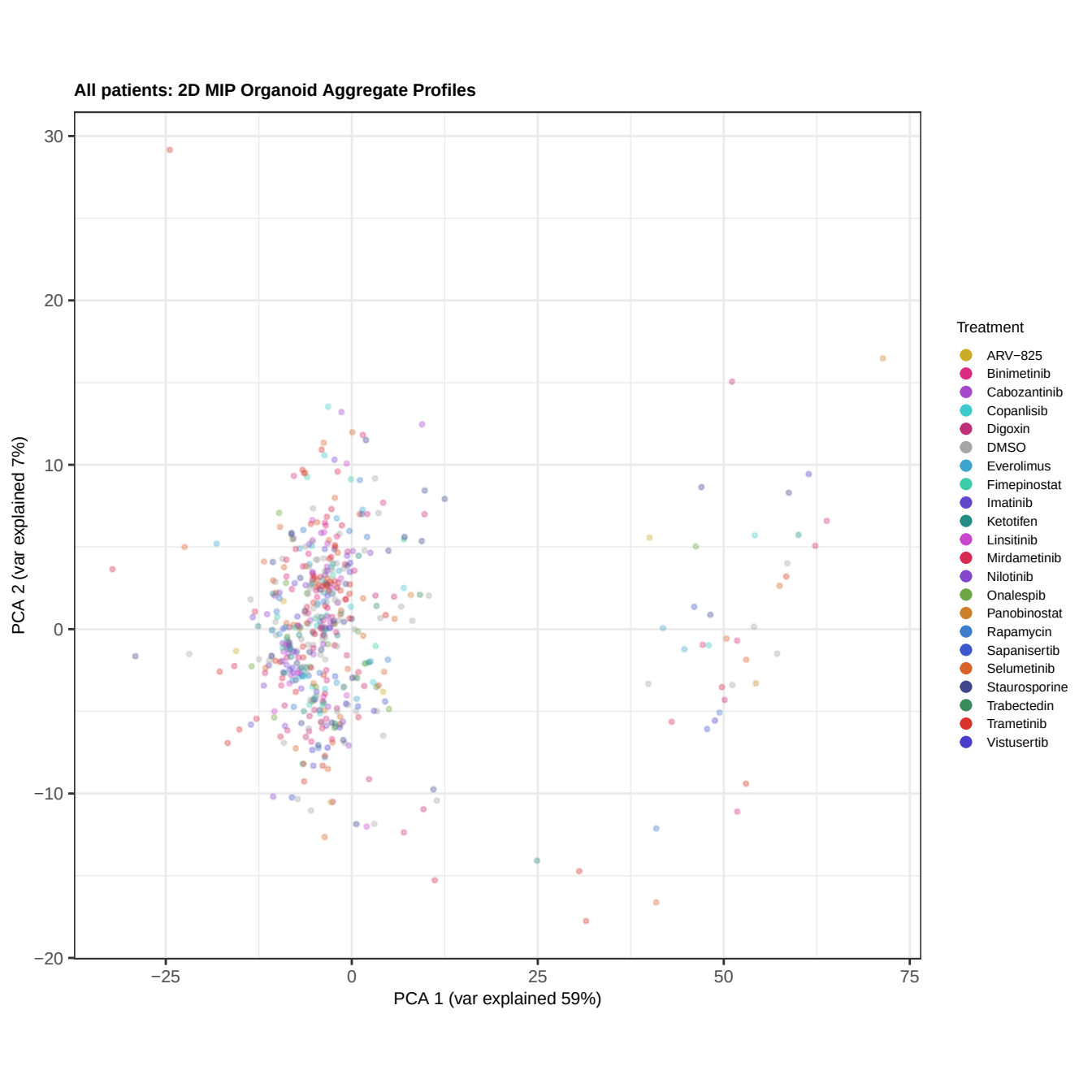

All patients: 2D MIP Single Cell Consensus Profiles
30
20
10
0
−10
−20
PCA 2 (var explained 12%)
0
25
50
75
PCA 1 (var explained 21%)
Tumor type
cNF
MPNST
Other
pNF

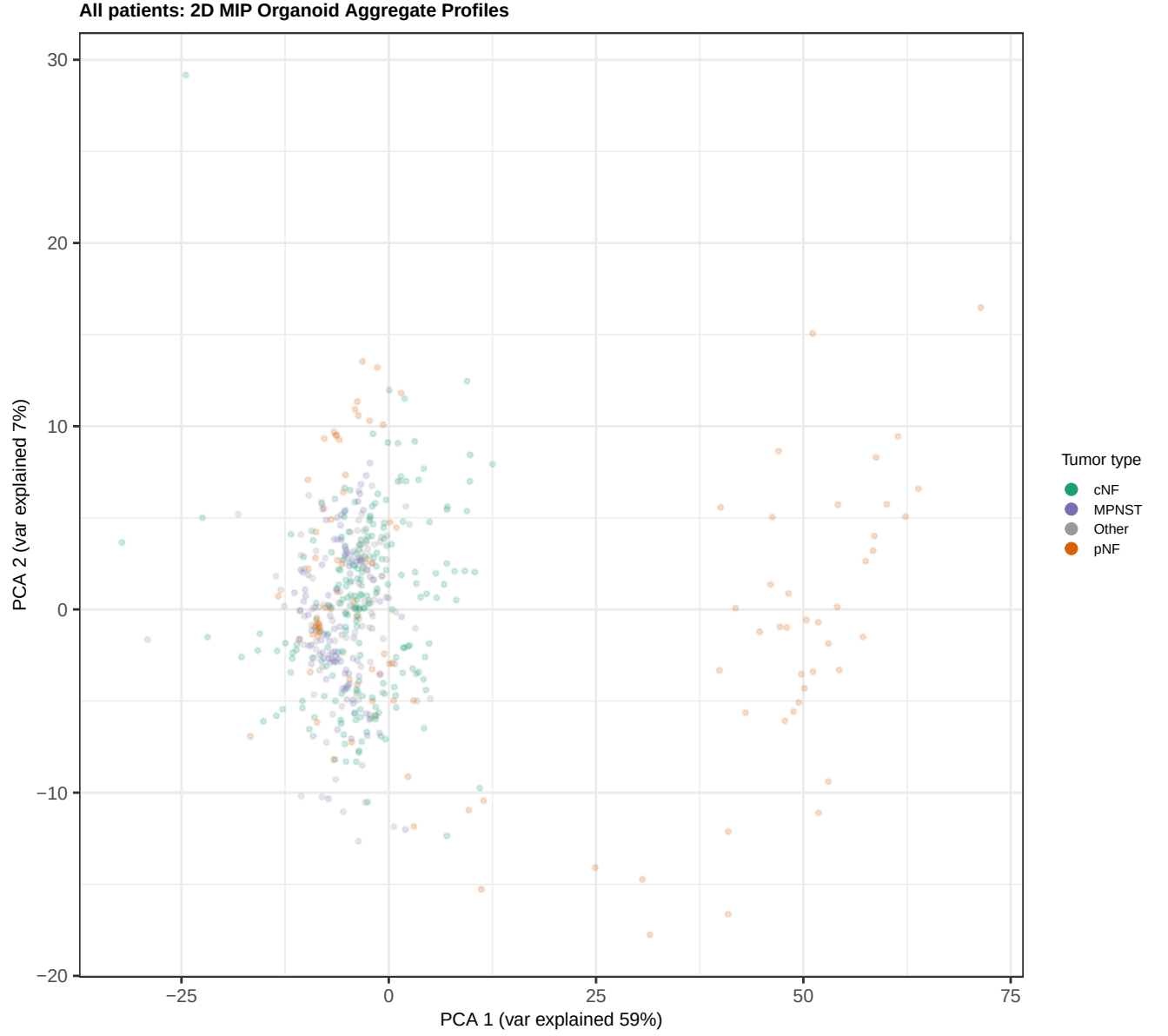

All patients: 2D MIP Organoid FS Profiles
100
0
−100
PCA 2 (var explained 5%)
0
50
100
150
PCA 1 (var explained 18%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

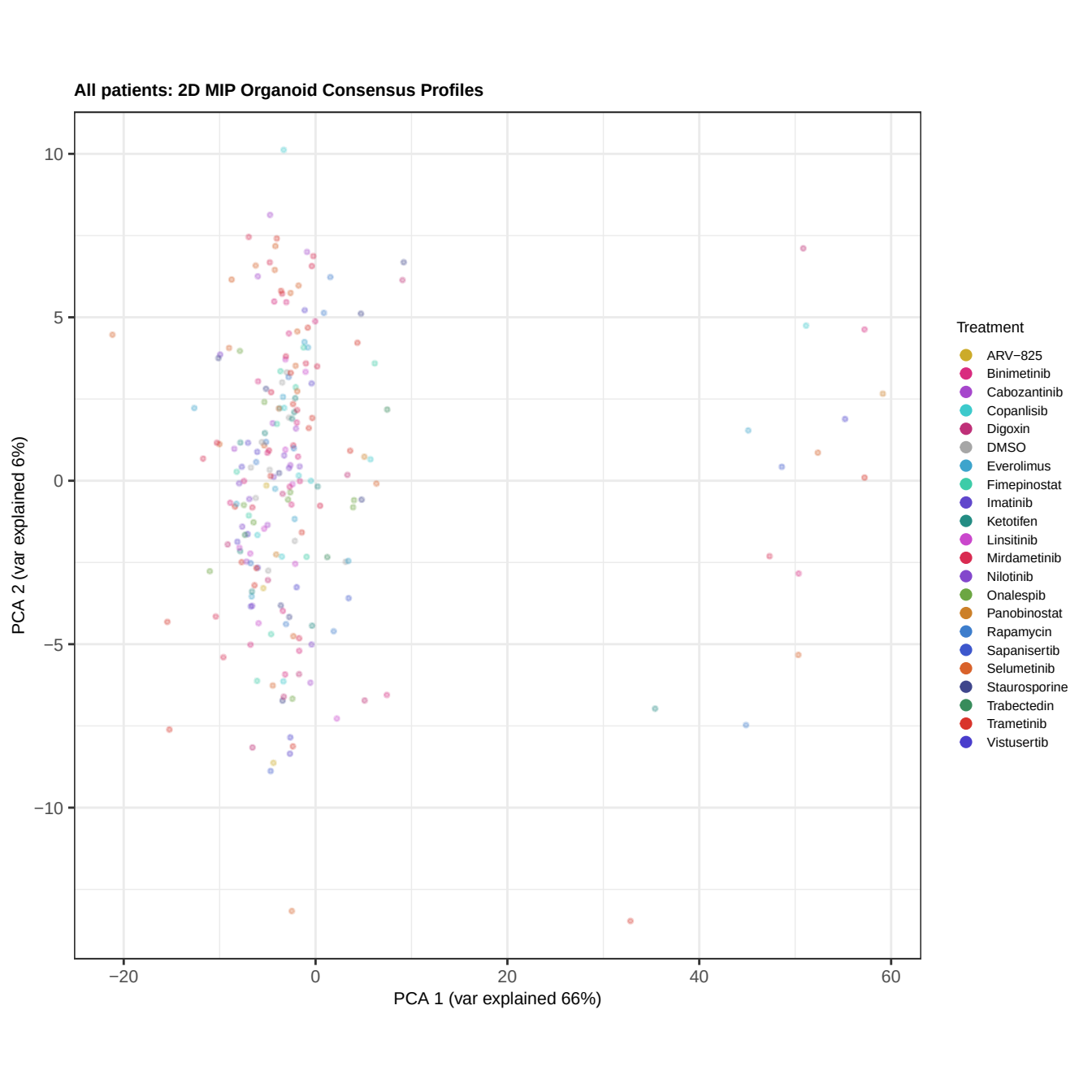

All patients: 2D MIP Organoid FS Profiles
100
0
−100
PCA 2 (var explained 5%)
0
50
100
150
PCA 1 (var explained 18%)
Tumor type
cNF
MPNST
Other
pNF

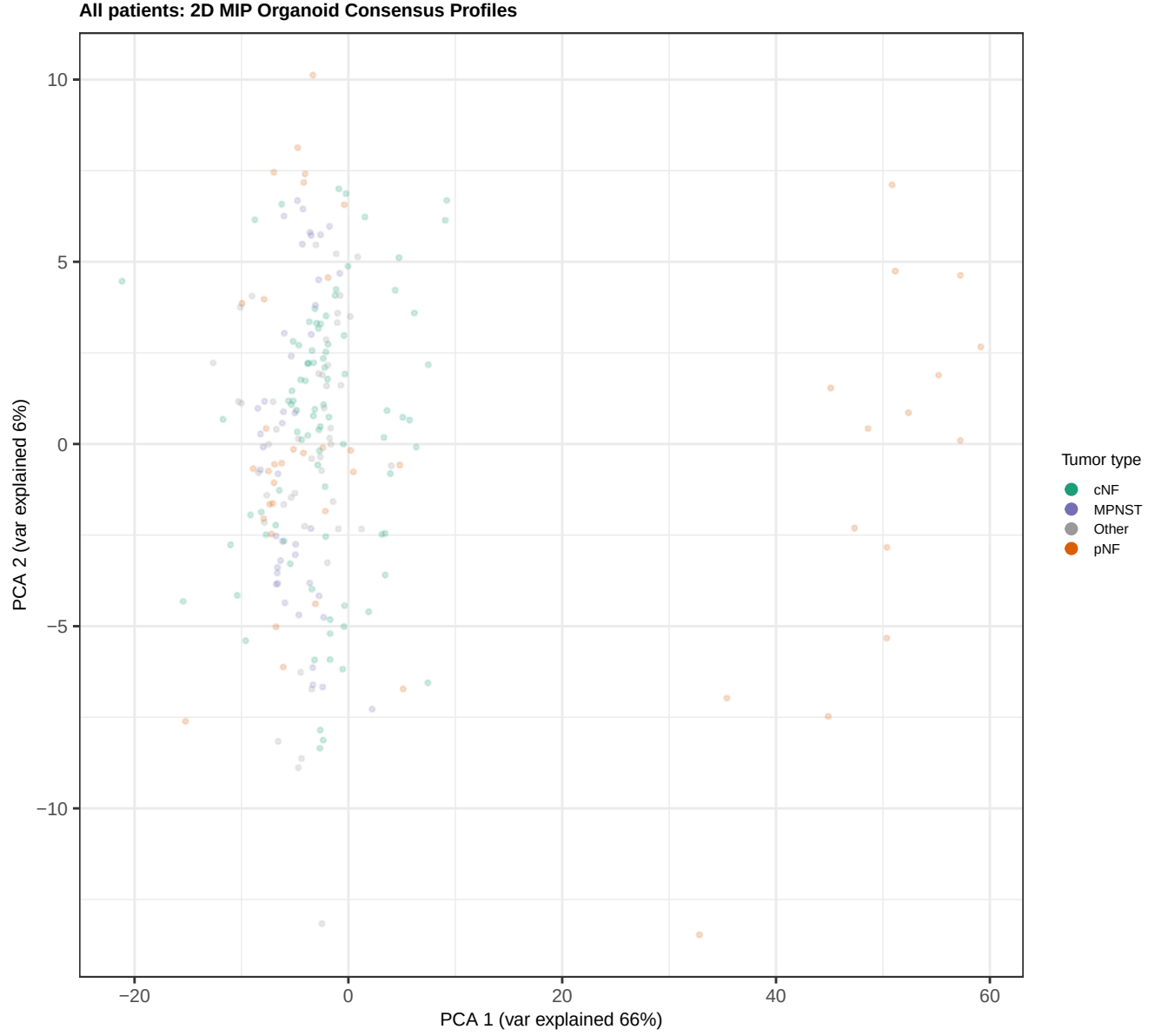

All patients: 2D MIP Organoid FS Profiles
NF0014_T1
NF0016_T1
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
100
0
−100
0
50
100
150
PCA 2 (var explained 5%)
PCA 1 (var explained 18%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

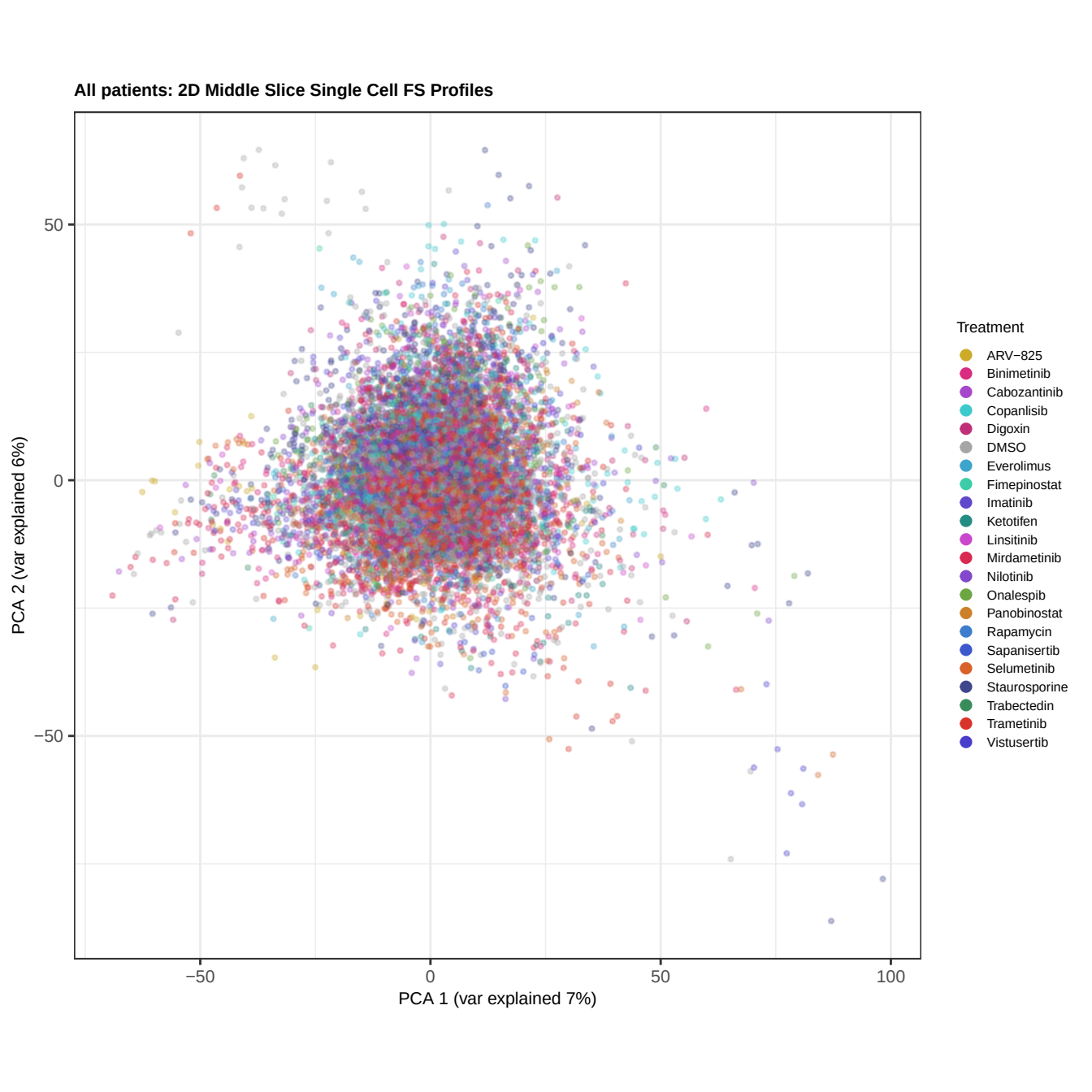

All patients: 2D MIP Organoid FS Profiles
cNF
MPNST
Other
pNF
100
0
−100
0
50
100
150
PCA 2 (var explained 5%)
PCA 1 (var explained 18%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

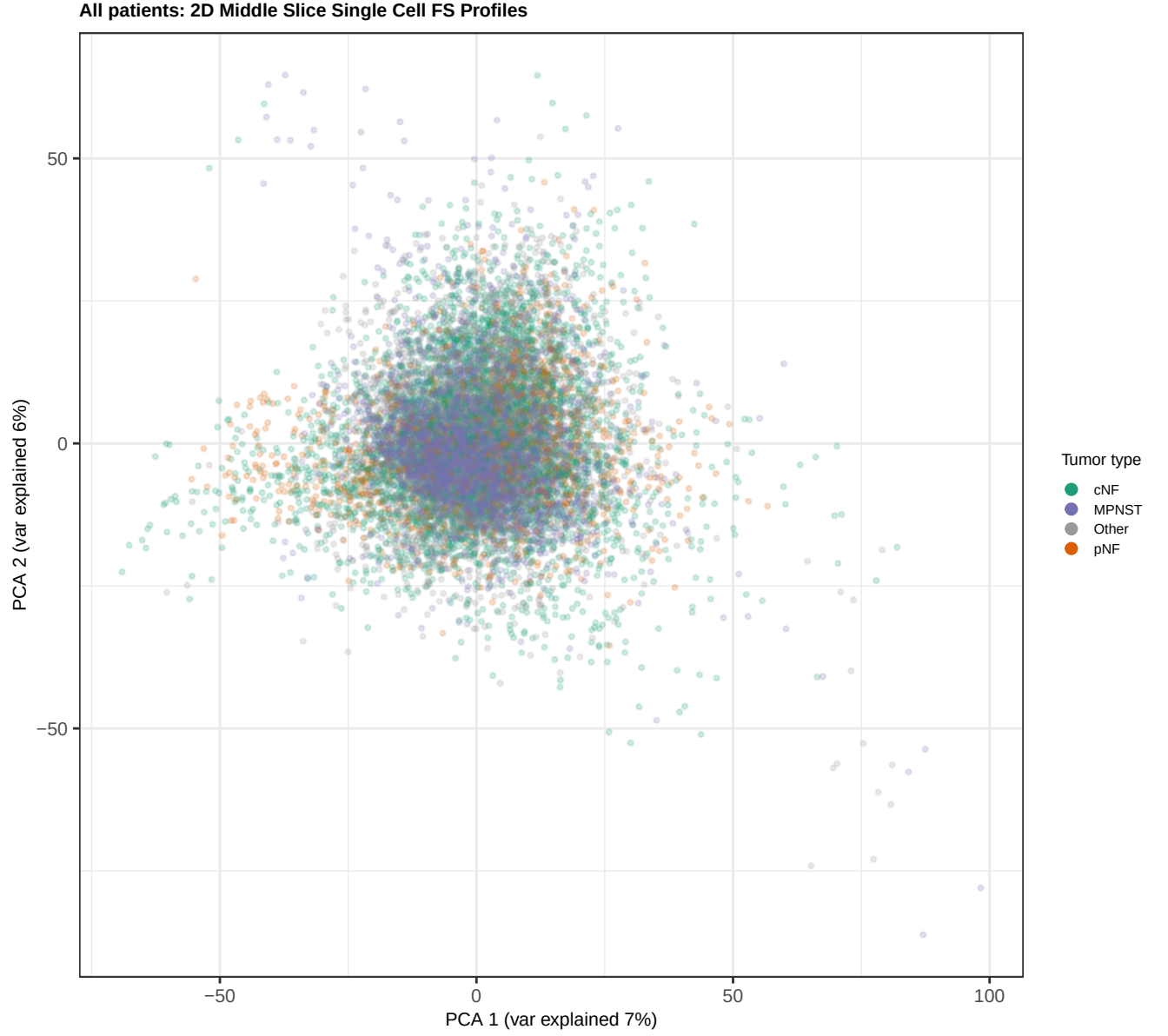

All patients: 2D MIP Organoid Aggregate Profiles
PCA 2 (var explained 7%)
PCA 1 (var explained 59%)
30
20
10
0
−10
−20
−25
0
25
50
75
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

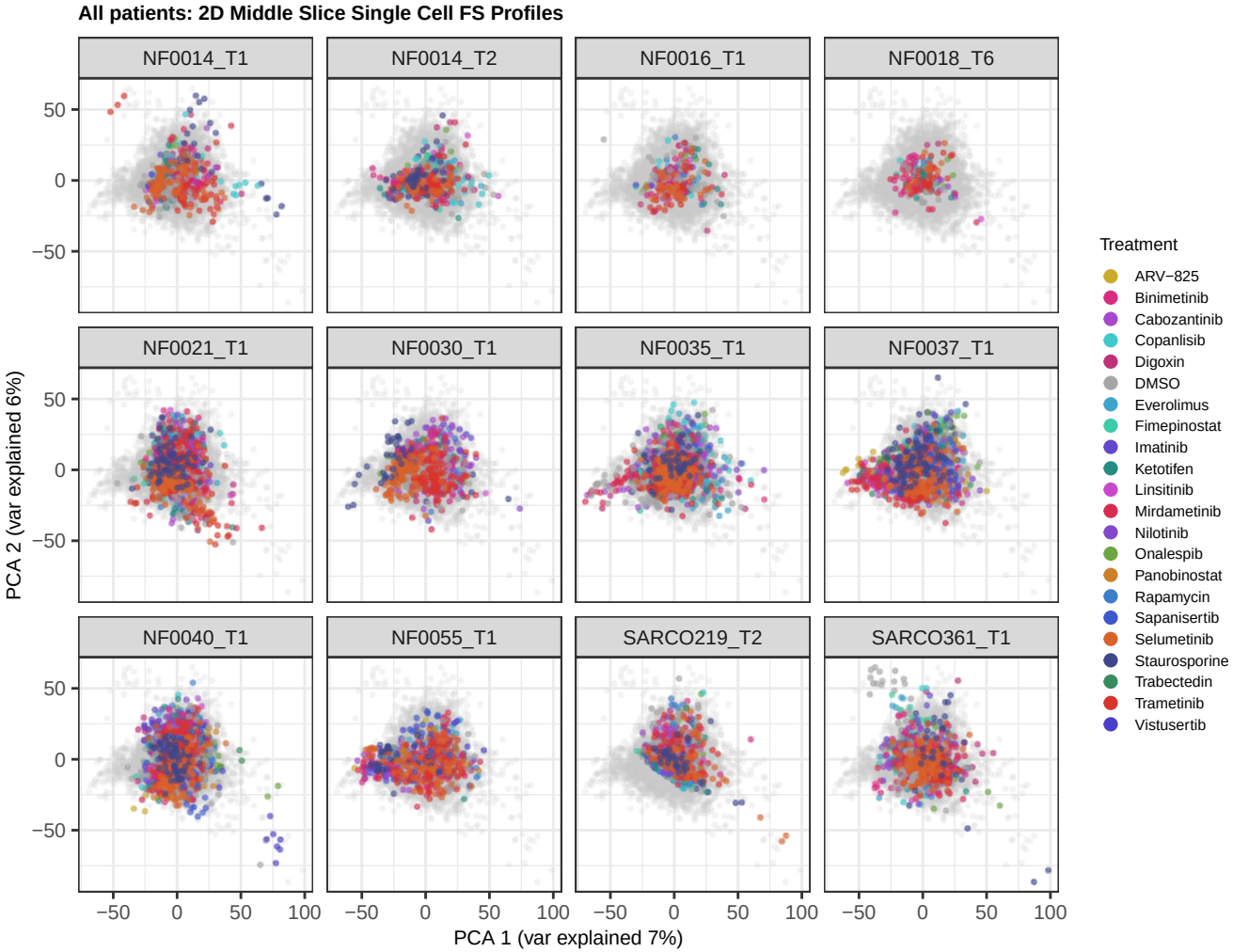

All patients: 2D MIP Organoid Aggregate Profiles
PCA 2 (var explained 7%)
30
20
10
0
−10
−20
−25
0
25
50
75
PCA 1 (var explained 59%)
Tumor type
cNF
MPNST
Other
pNF

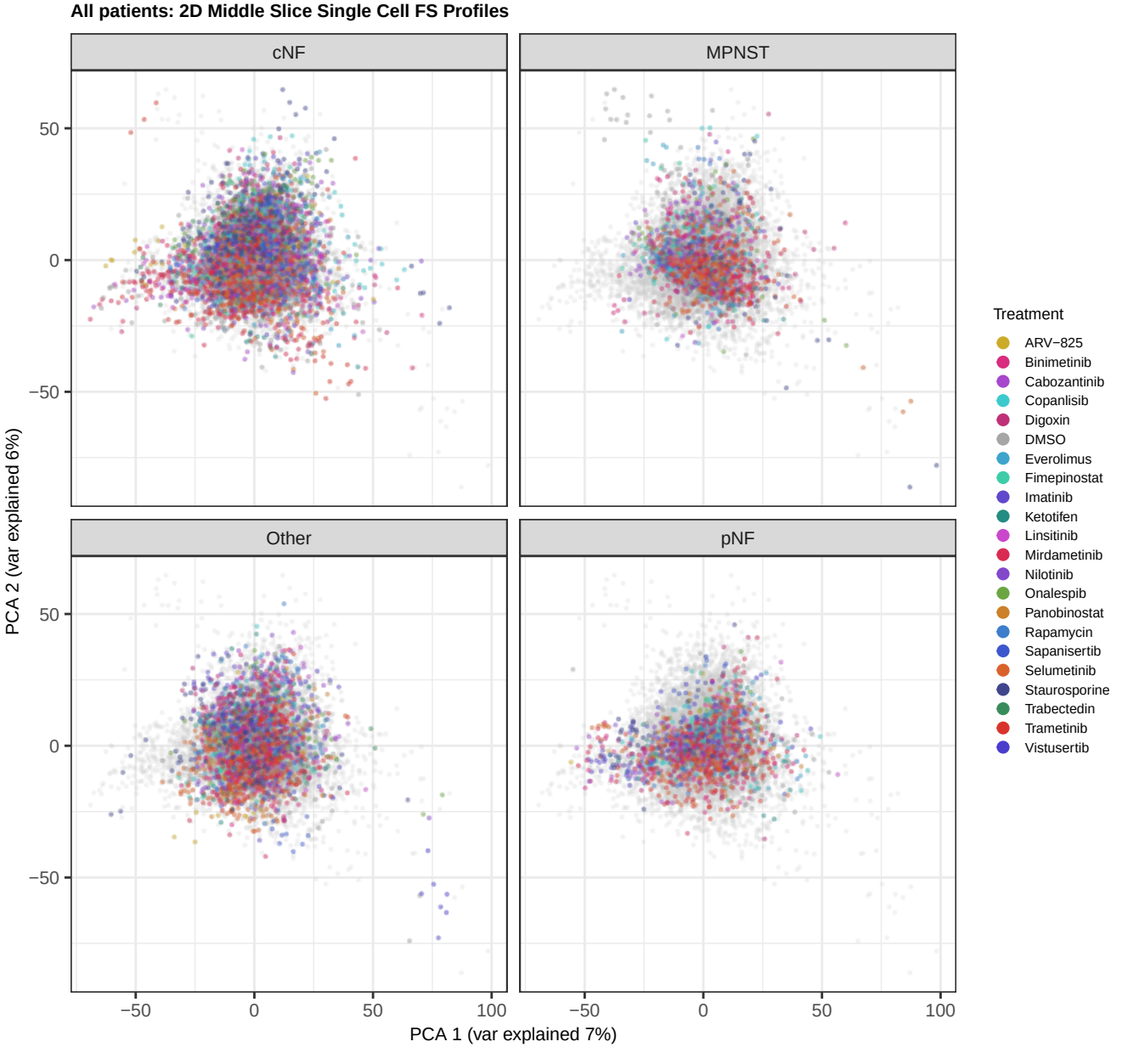

All patients: 2D MIP Organoid Consensus Profiles
PCA 2 (var explained 6%)
PCA 1 (var explained 66%)
10
5
0
−5
−10
−20
0
20
40
60
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

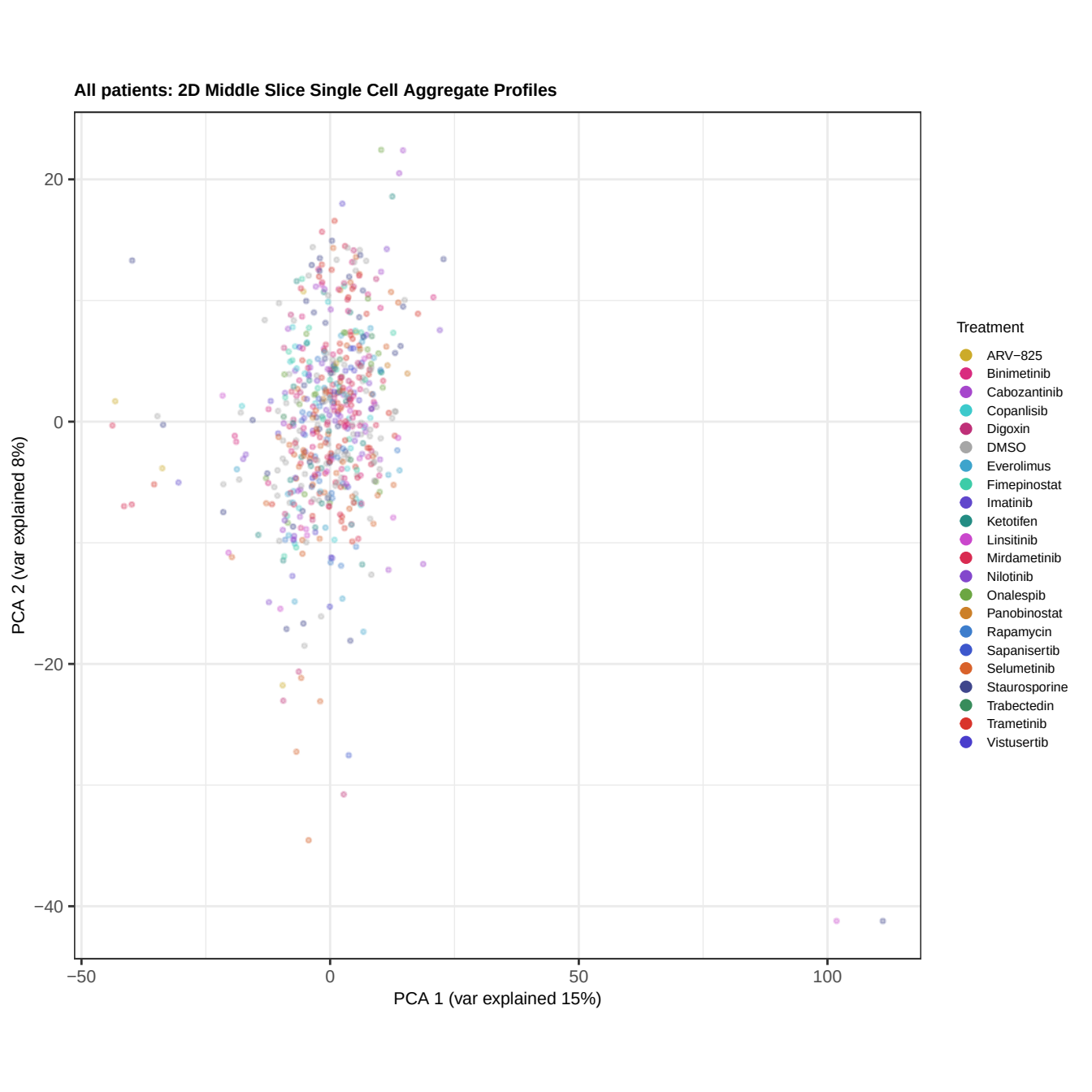

All patients: 2D MIP Organoid Consensus Profiles
10
5
0
−5
−10
PCA 2 (var explained 6%)
−20
0
20
40
60
PCA 1 (var explained 66%)
Tumor type
cNF
MPNST
Other
pNF

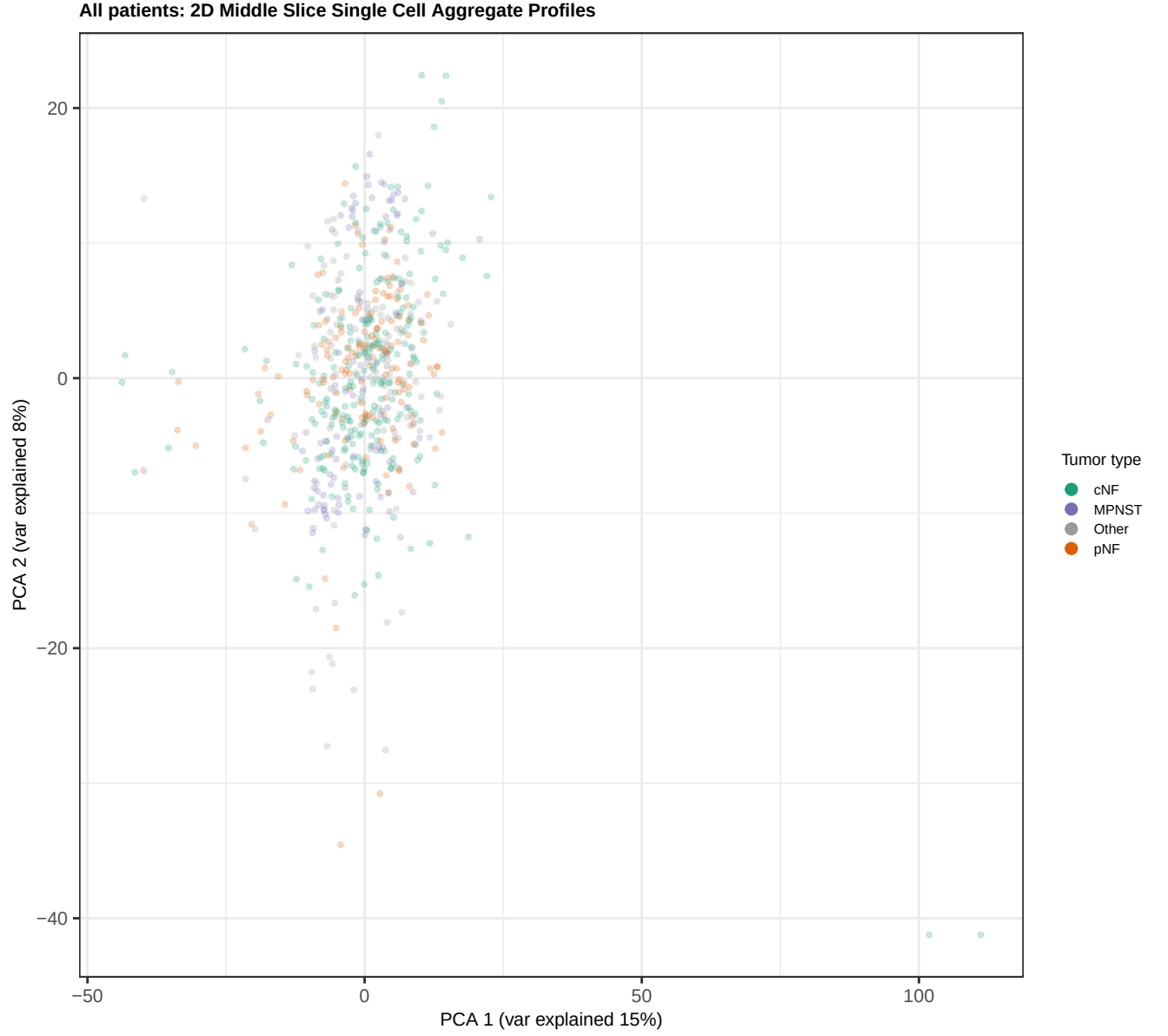

All patients: 2D Middle Slice Single Cell FS Profiles
PCA 2 (var explained 6%)
50
0
−50
−50
0
50
100
PCA 1 (var explained 7%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

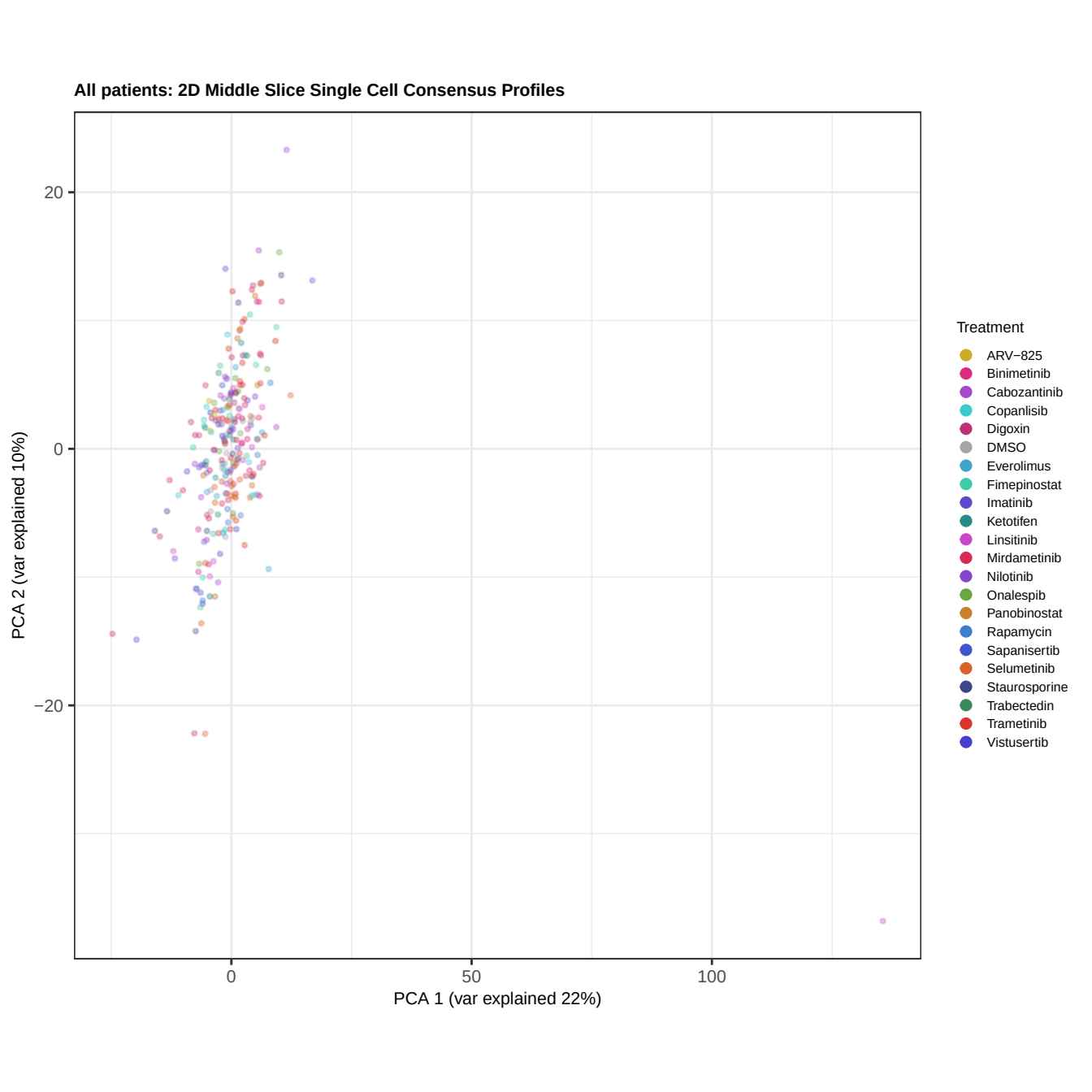

All patients: 2D Middle Slice Single Cell FS Profiles
50
0
−50
PCA 2 (var explained 6%)
−50
0
50
100
PCA 1 (var explained 7%)
Tumor type
cNF
MPNST
Other
pNF

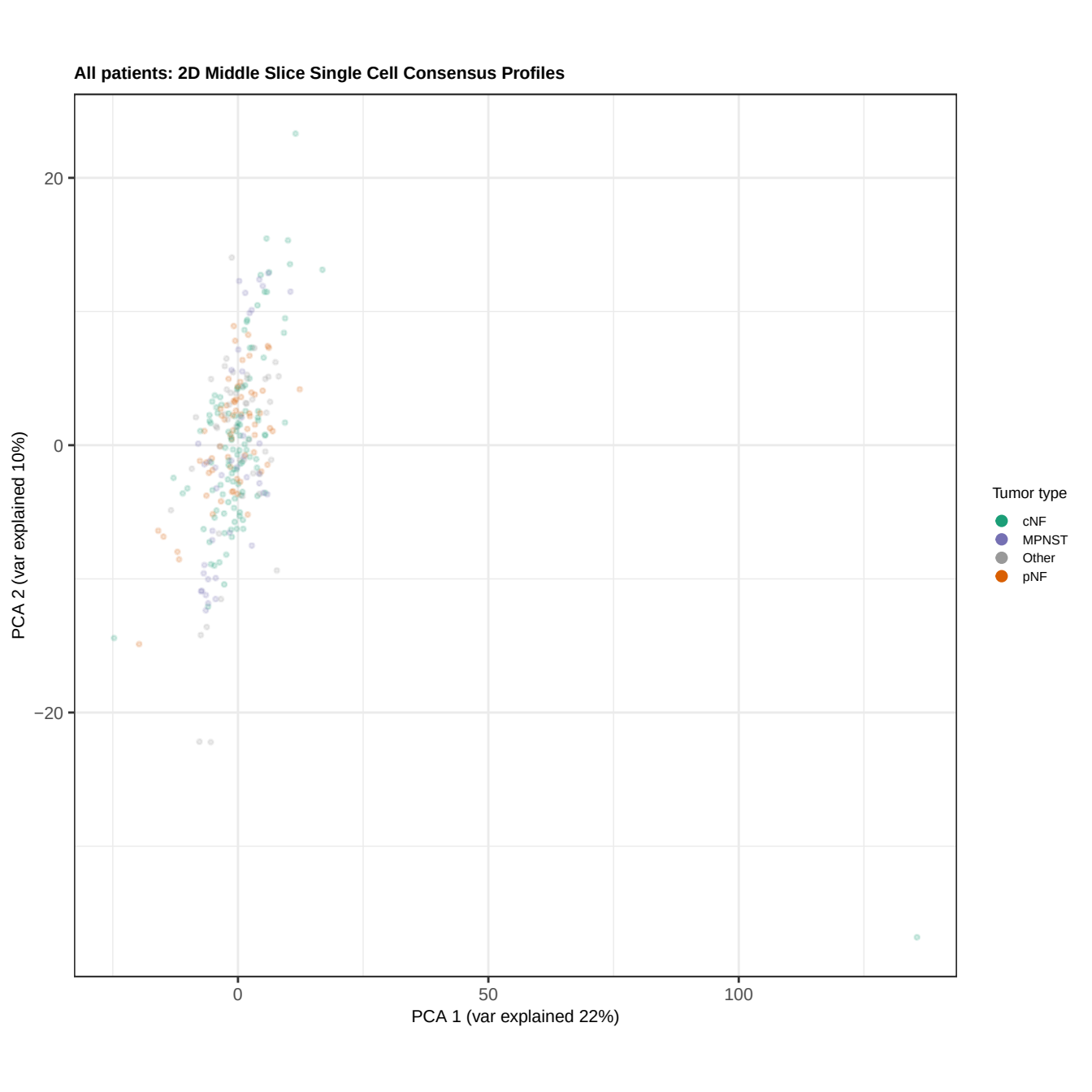

All patients: 2D Middle Slice Single Cell FS Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
PCA 2 (var explained 6%)
PCA 1 (var explained 7%)
50
0
−50
100
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

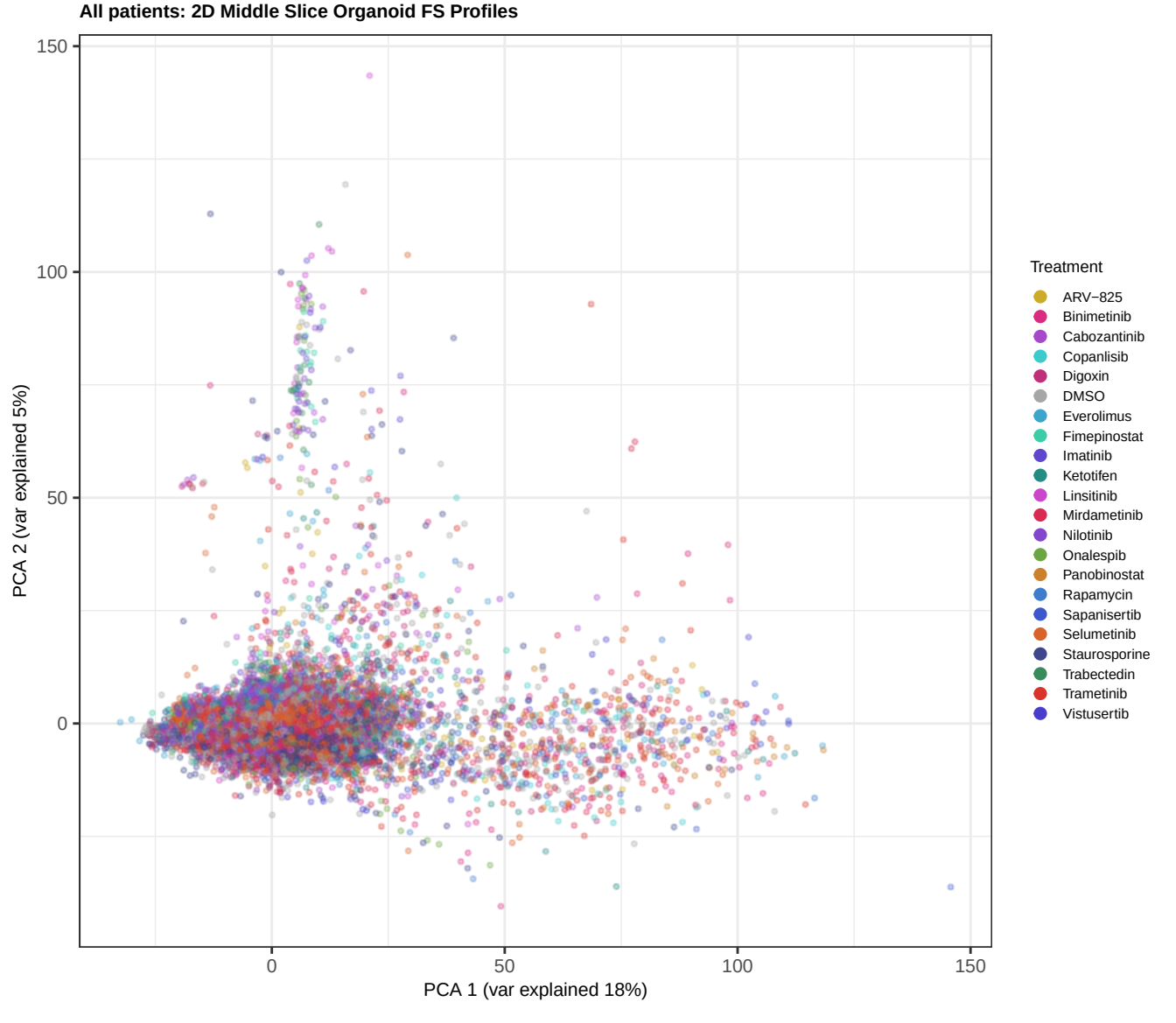

All patients: 2D Middle Slice Single Cell FS Profiles
cNF
MPNST
Other
pNF
PCA 2 (var explained 6%)
PCA 1 (var explained 7%)
50
0
−50
100
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib


All patients: 2D Middle Slice Single Cell Aggregate Profiles
20
0
−20
−40
PCA 2 (var explained 8%)
−50
0
50
100
PCA 1 (var explained 15%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

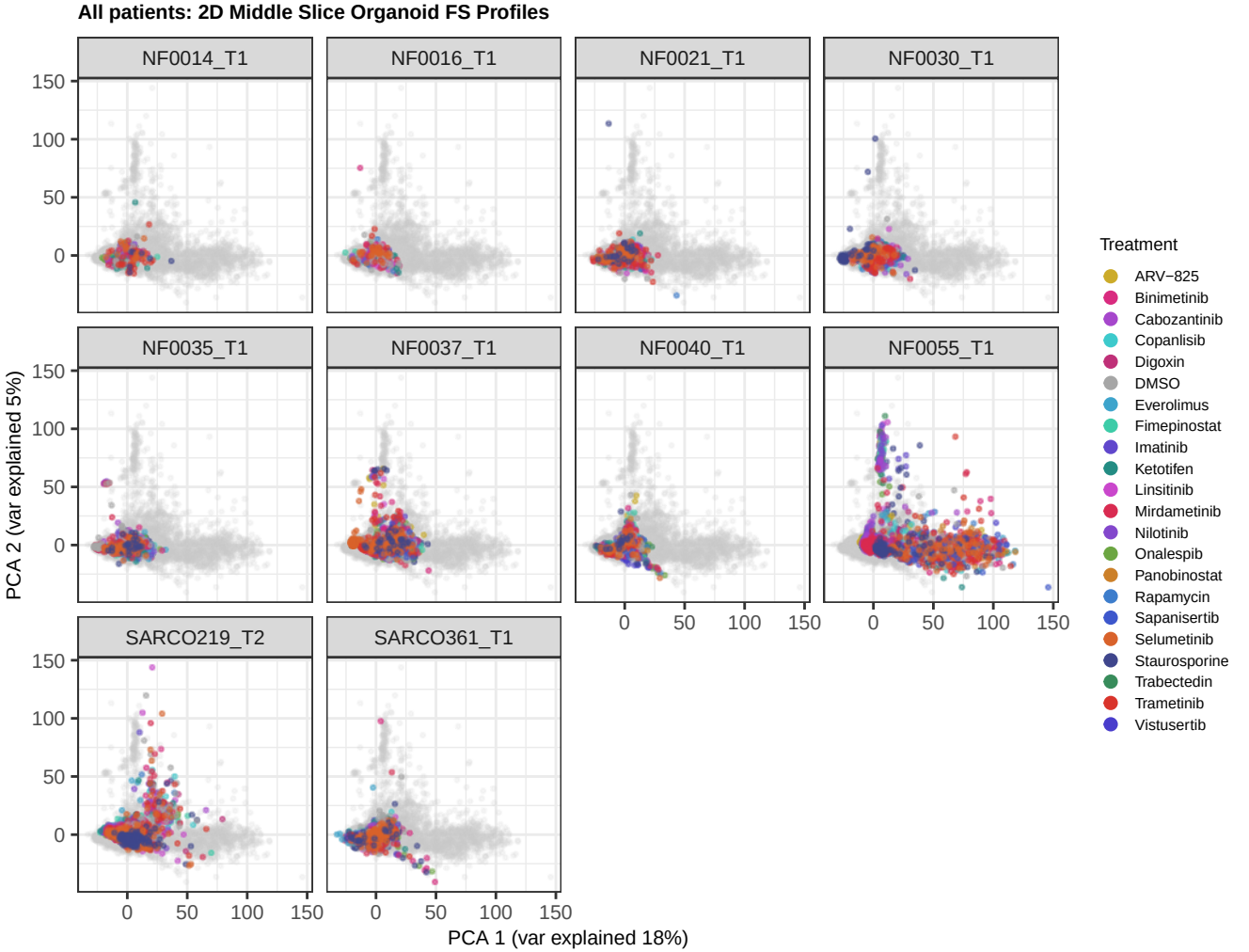

All patients: 2D Middle Slice Single Cell Aggregate Profiles
20
0
−20
−40
PCA 2 (var explained 8%)
−50
0
50
100
PCA 1 (var explained 15%)
Tumor type
cNF
MPNST
Other
pNF

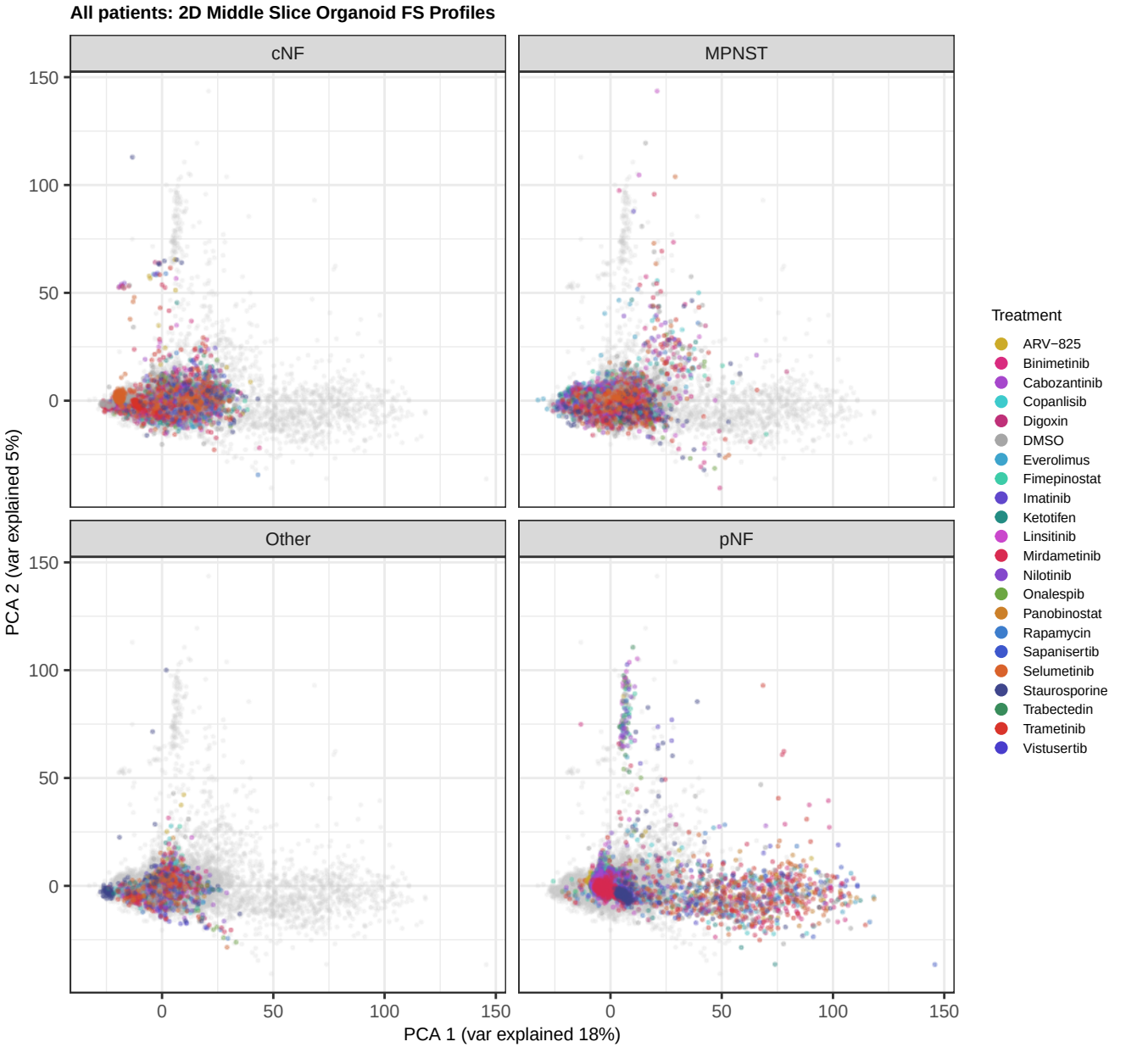

All patients: 2D Middle Slice Single Cell Consensus Profiles
20
0
−20
PCA 2 (var explained 10%)
0
50
100
PCA 1 (var explained 22%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

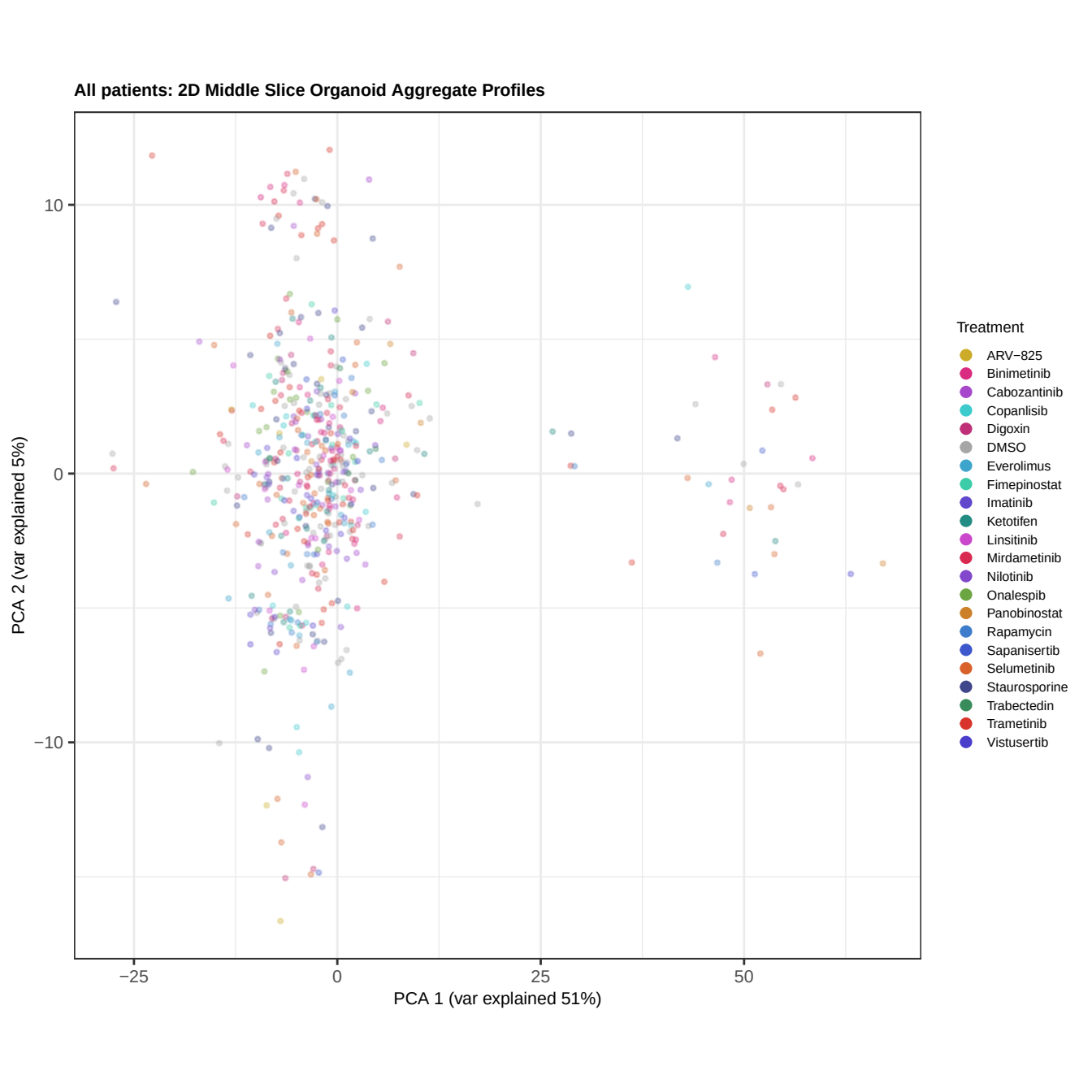

All patients: 2D Middle Slice Single Cell Consensus Profiles
20
0
−20
PCA 2 (var explained 10%)
0
50
100
PCA 1 (var explained 22%)
Tumor type
cNF
MPNST
Other
pNF

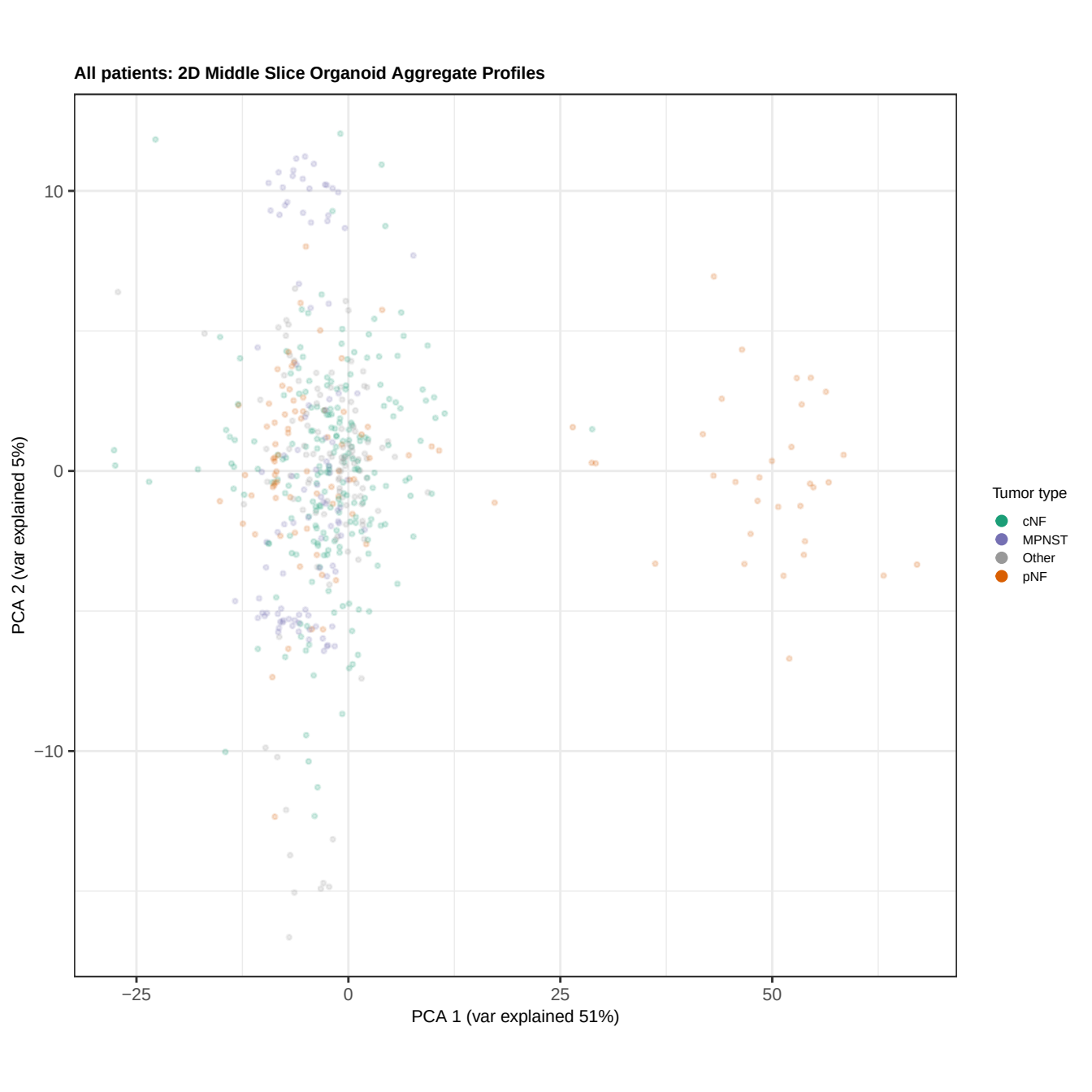

All patients: 2D Middle Slice Organoid FS Profiles
150
100
50
0
PCA 2 (var explained 5%)
0
50
100
150
PCA 1 (var explained 18%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

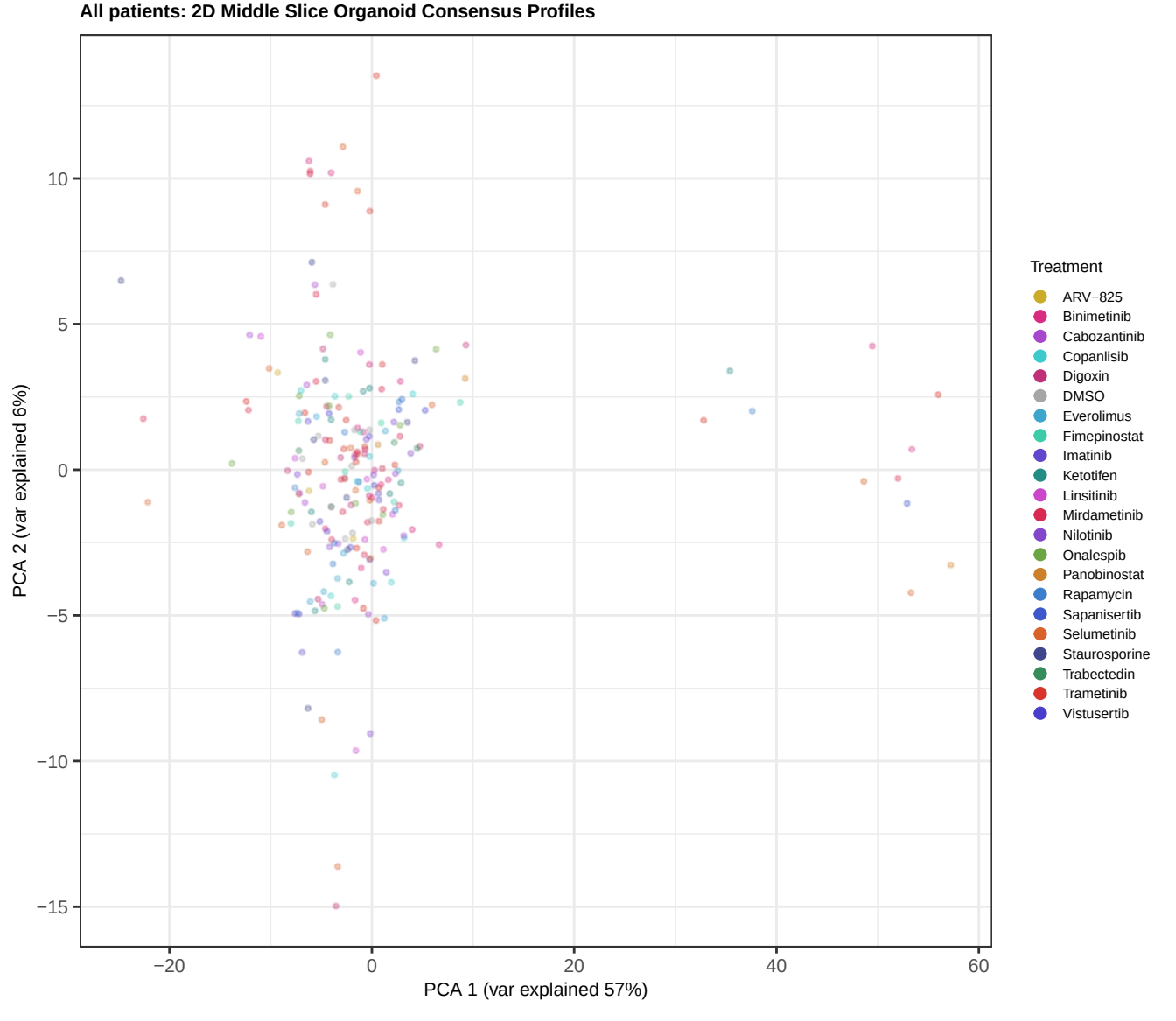

All patients: 2D Middle Slice Organoid FS Profiles
150
100
50
0
PCA 2 (var explained 5%)
0
50
100
150
PCA 1 (var explained 18%)
Tumor type
cNF
MPNST
Other
pNF

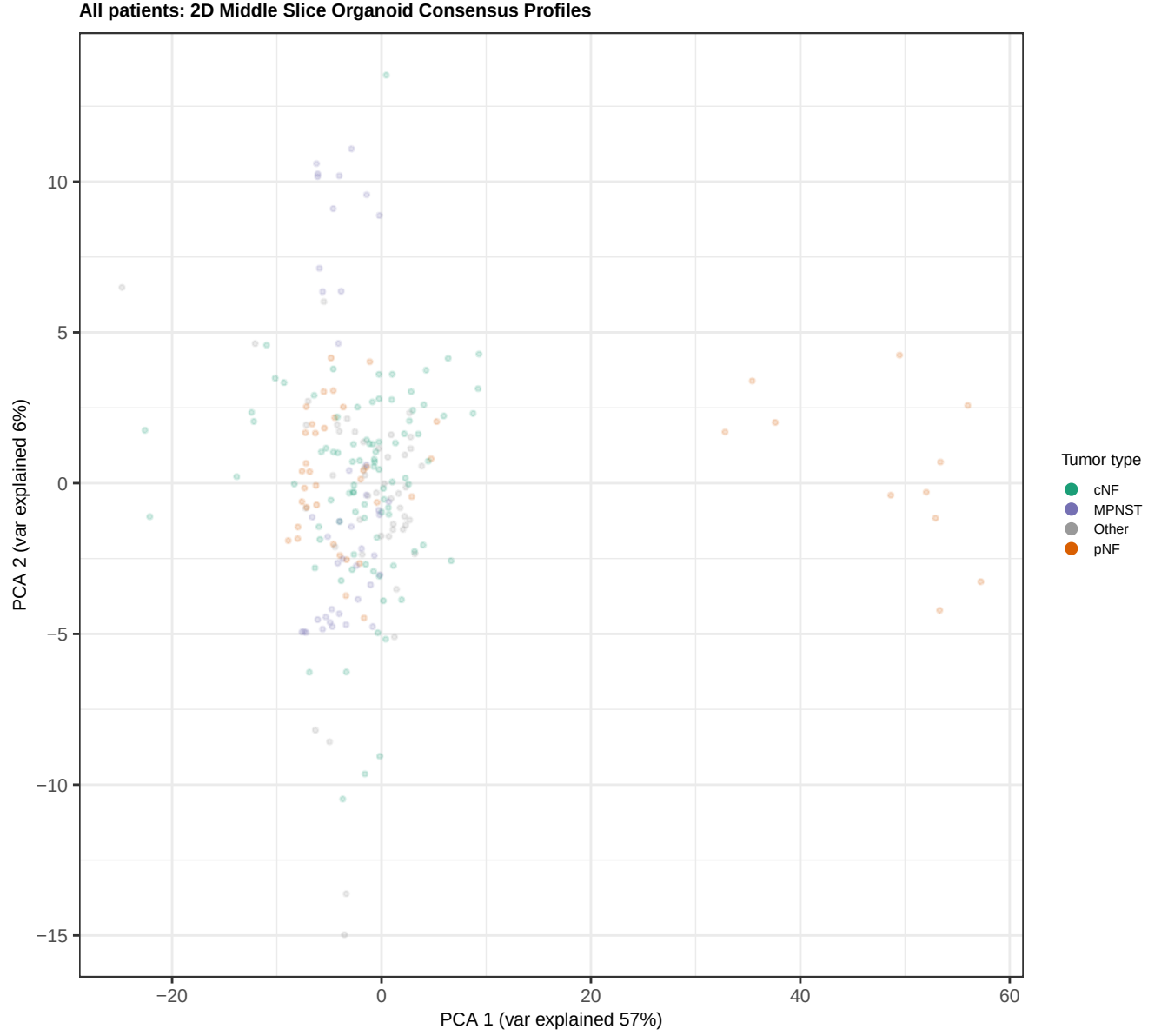

All patients: 2D Middle Slice Organoid FS Profiles
NF0014_T1
NF0016_T1
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
PCA 2 (var explained 5%)
PCA 1 (var explained 18%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

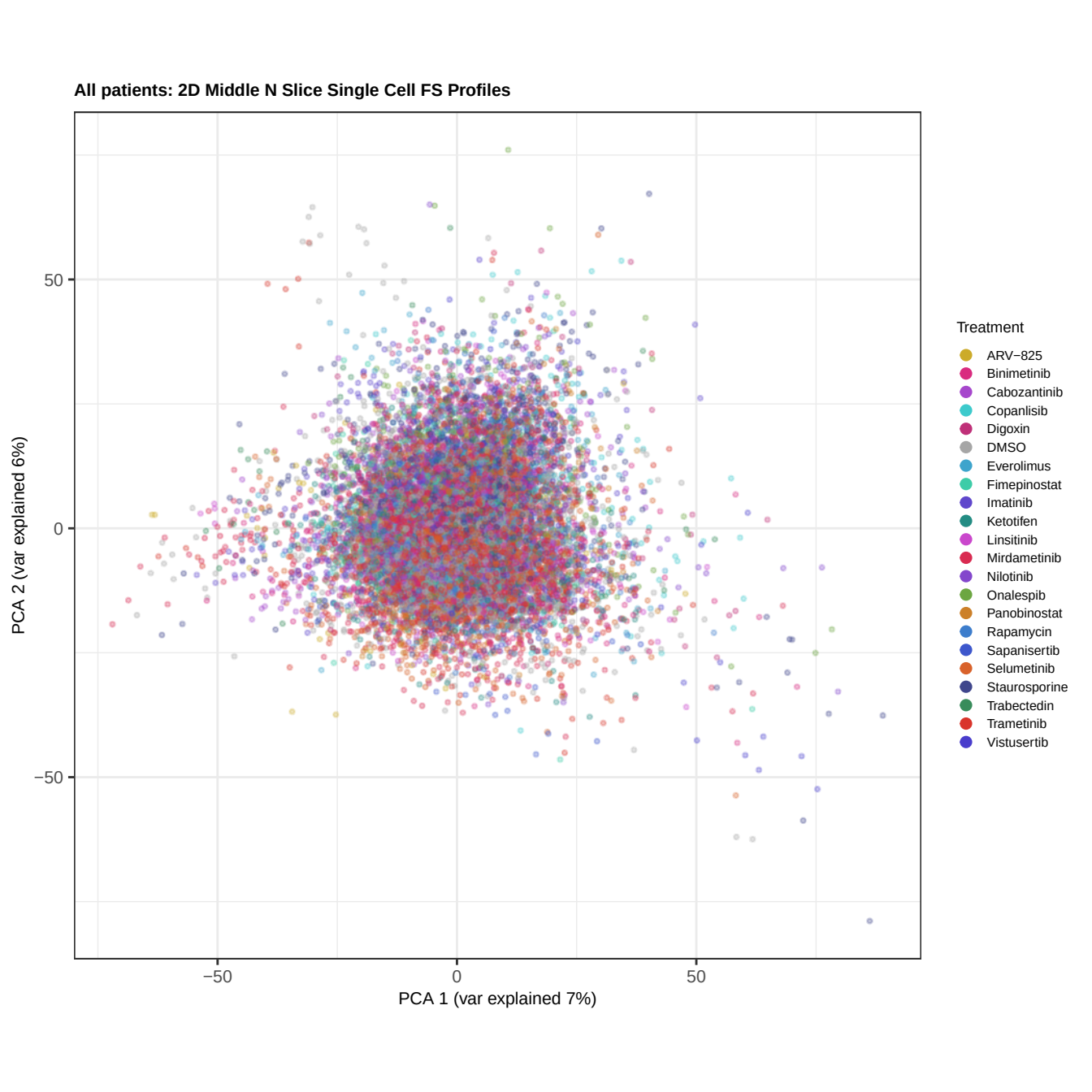

All patients: 2D Middle Slice Organoid FS Profiles
cNF
MPNST
Other
pNF
PCA 2 (var explained 5%)
PCA 1 (var explained 18%)
150
100
50
0
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

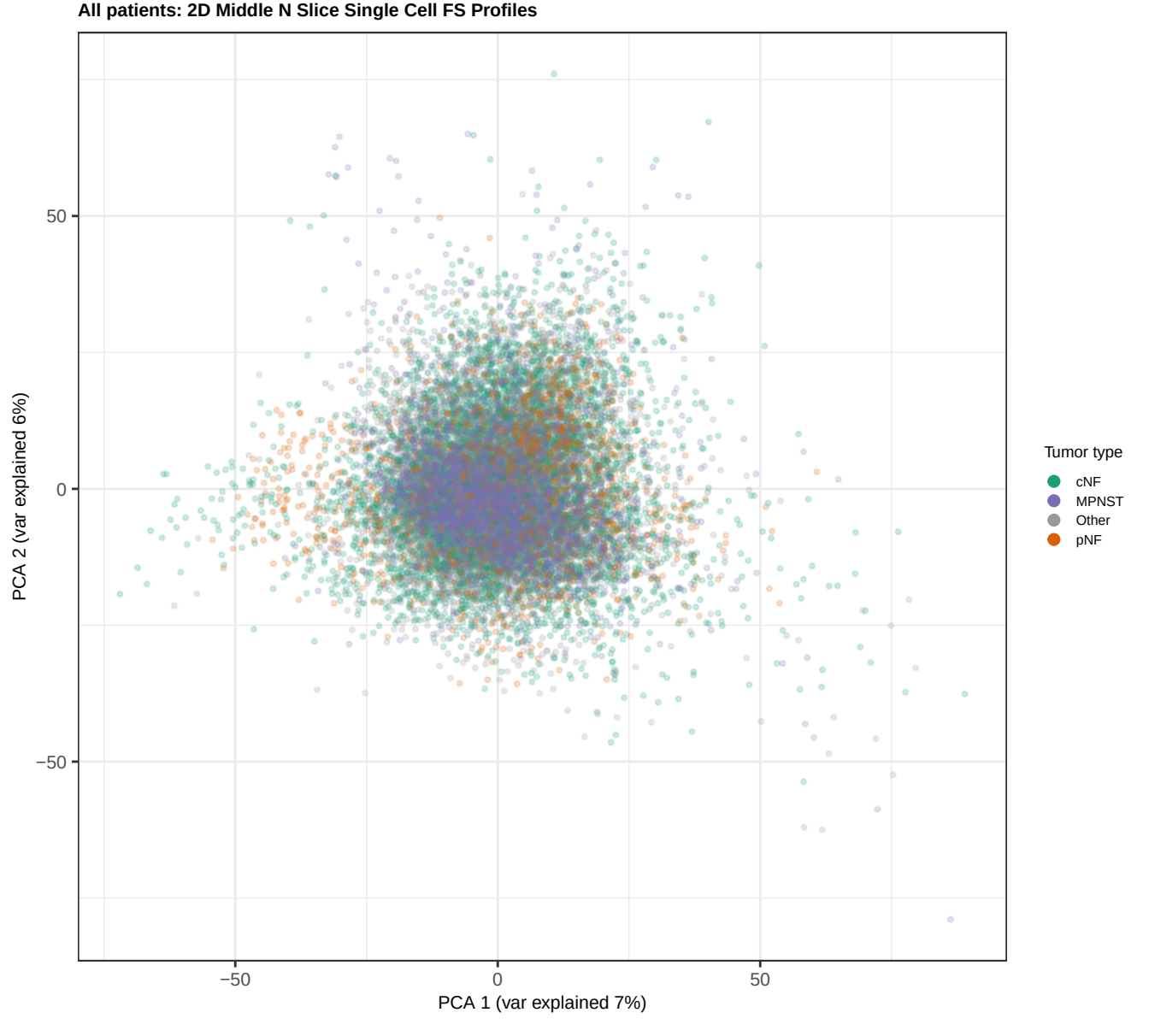

All patients: 2D Middle Slice Organoid Aggregate Profiles
PCA 2 (var explained 5%)
10
0
−10
−25
0
25
50
PCA 1 (var explained 51%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

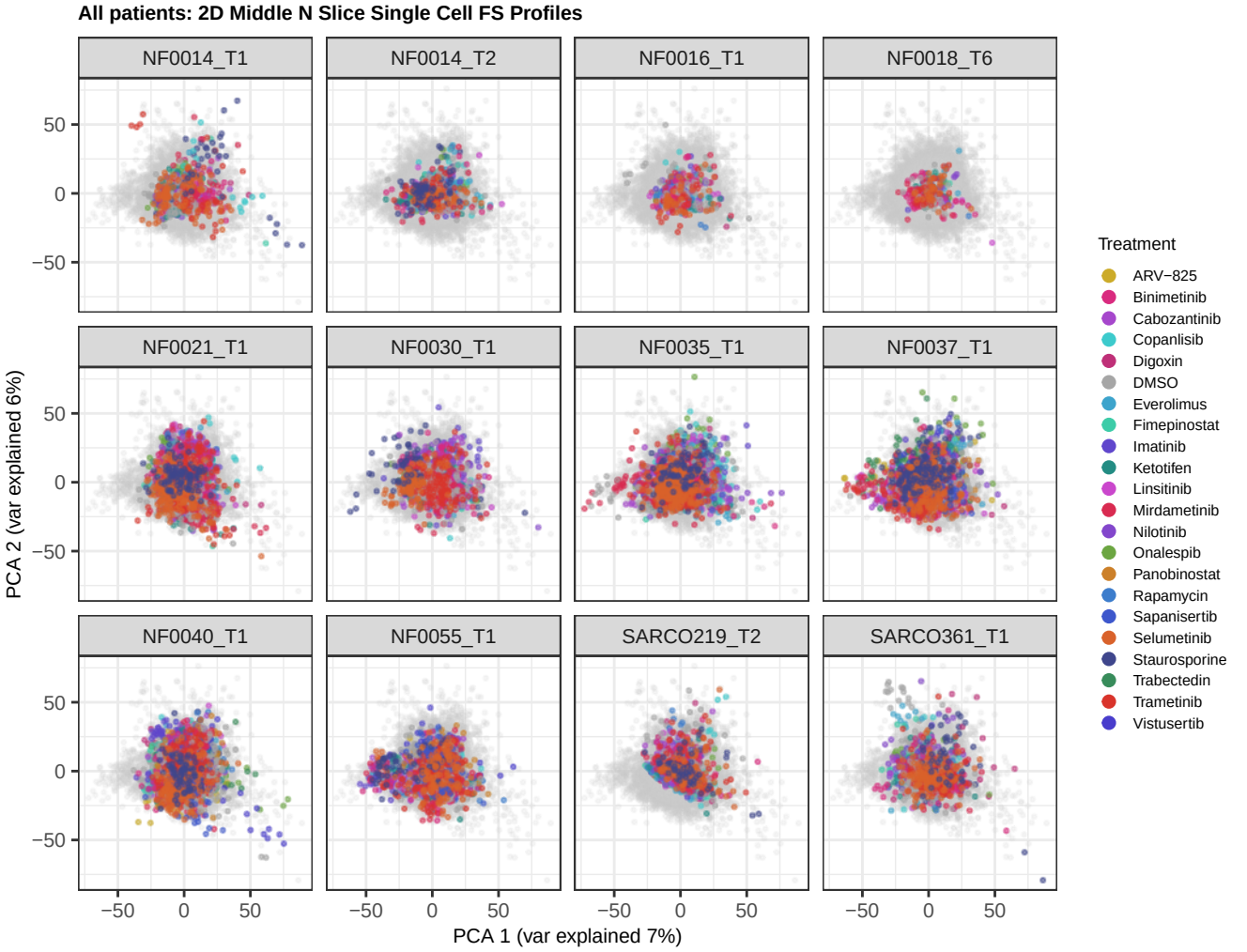

All patients: 2D Middle Slice Organoid Aggregate Profiles
PCA 2 (var explained 5%)
10
0
−10
−25
0
25
50
PCA 1 (var explained 51%)
Tumor type
cNF
MPNST
Other
pNF

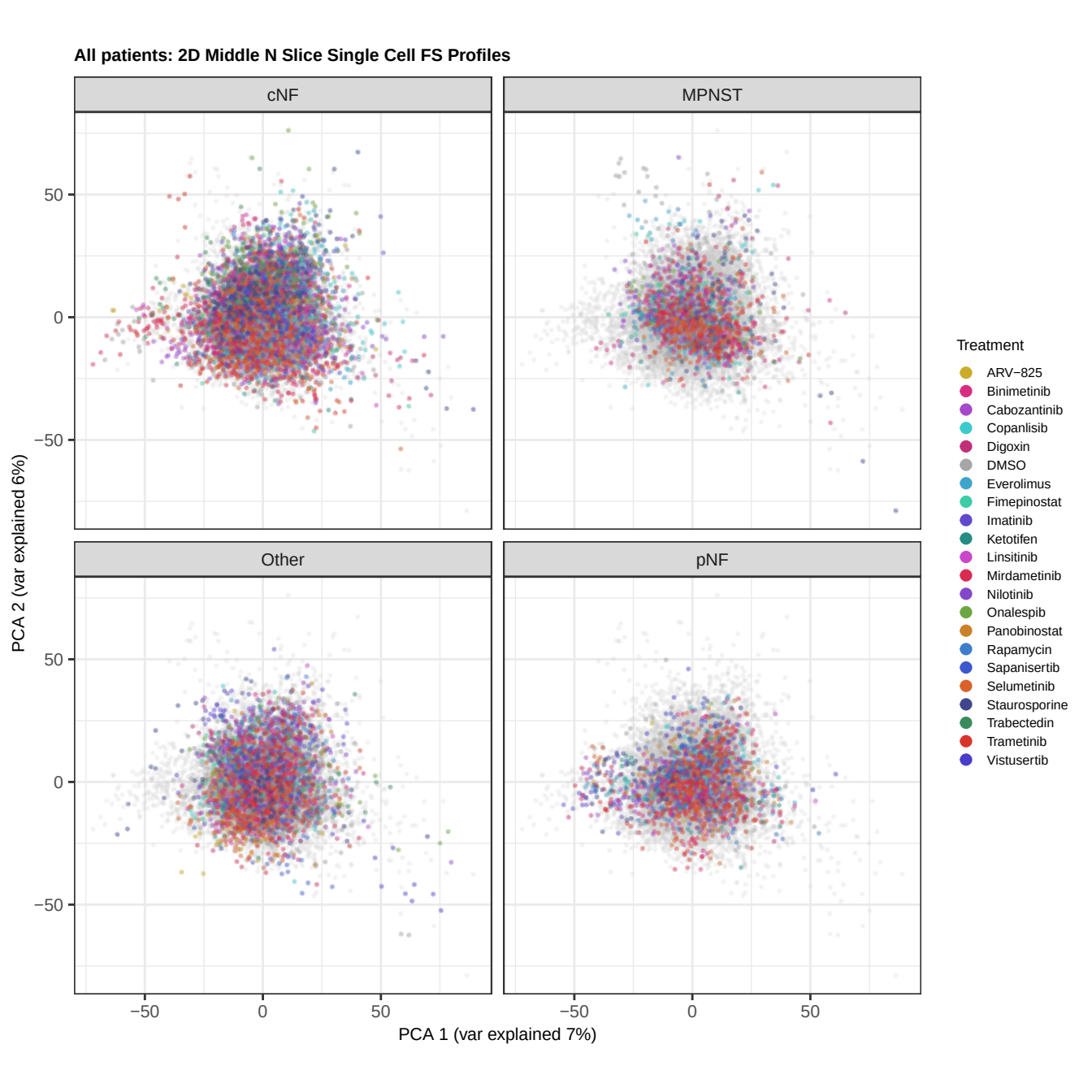

All patients: 2D Middle Slice Organoid Consensus Profiles
PCA 2 (var explained 6%)
10
5
0
−5
−10
−15
−20
0
20
40
60
PCA 1 (var explained 57%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

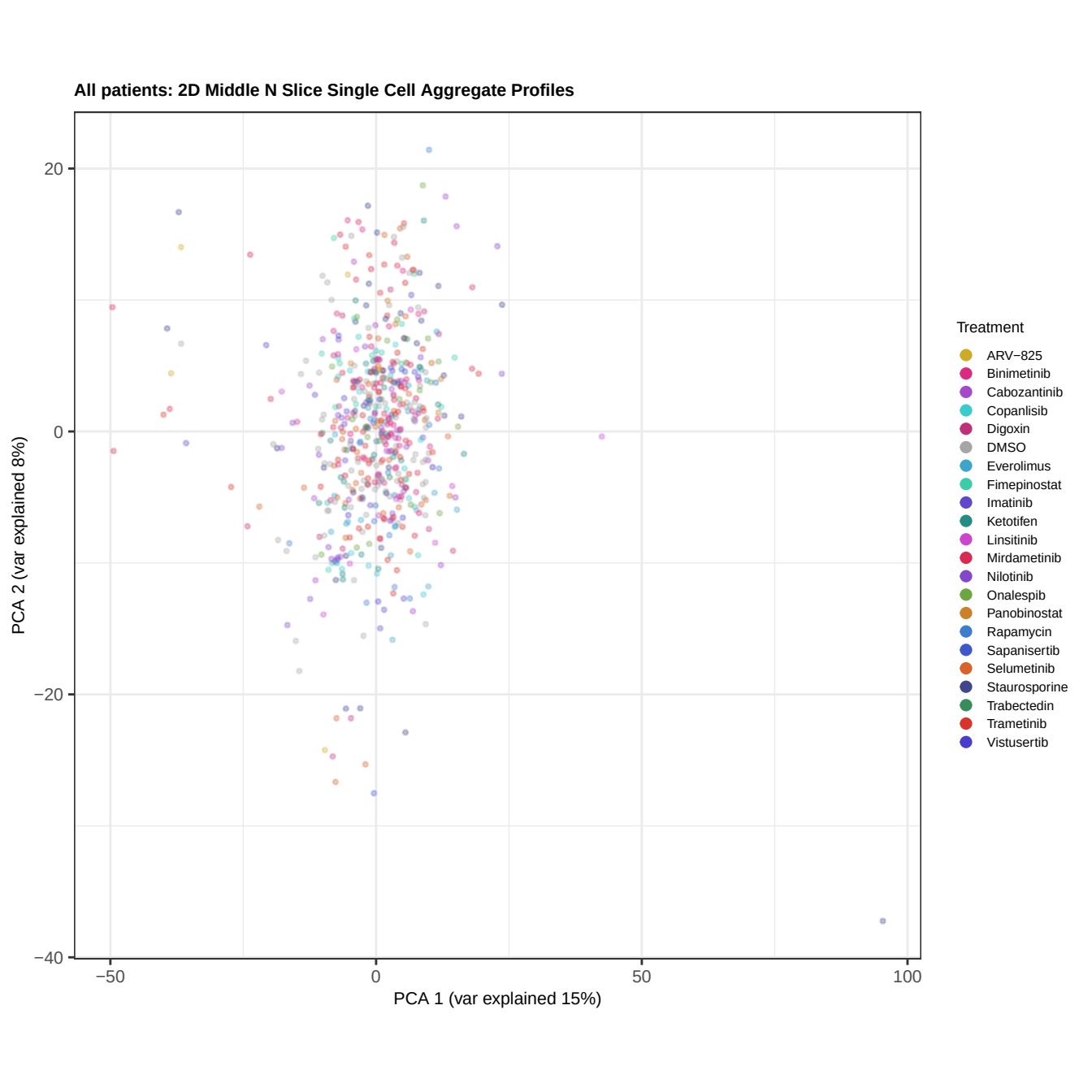

All patients: 2D Middle Slice Organoid Consensus Profiles
PCA 2 (var explained 6%)
10
5
0
−5
−10
−15
−20
0
20
40
60
PCA 1 (var explained 57%)
Tumor type
cNF
MPNST
Other
pNF

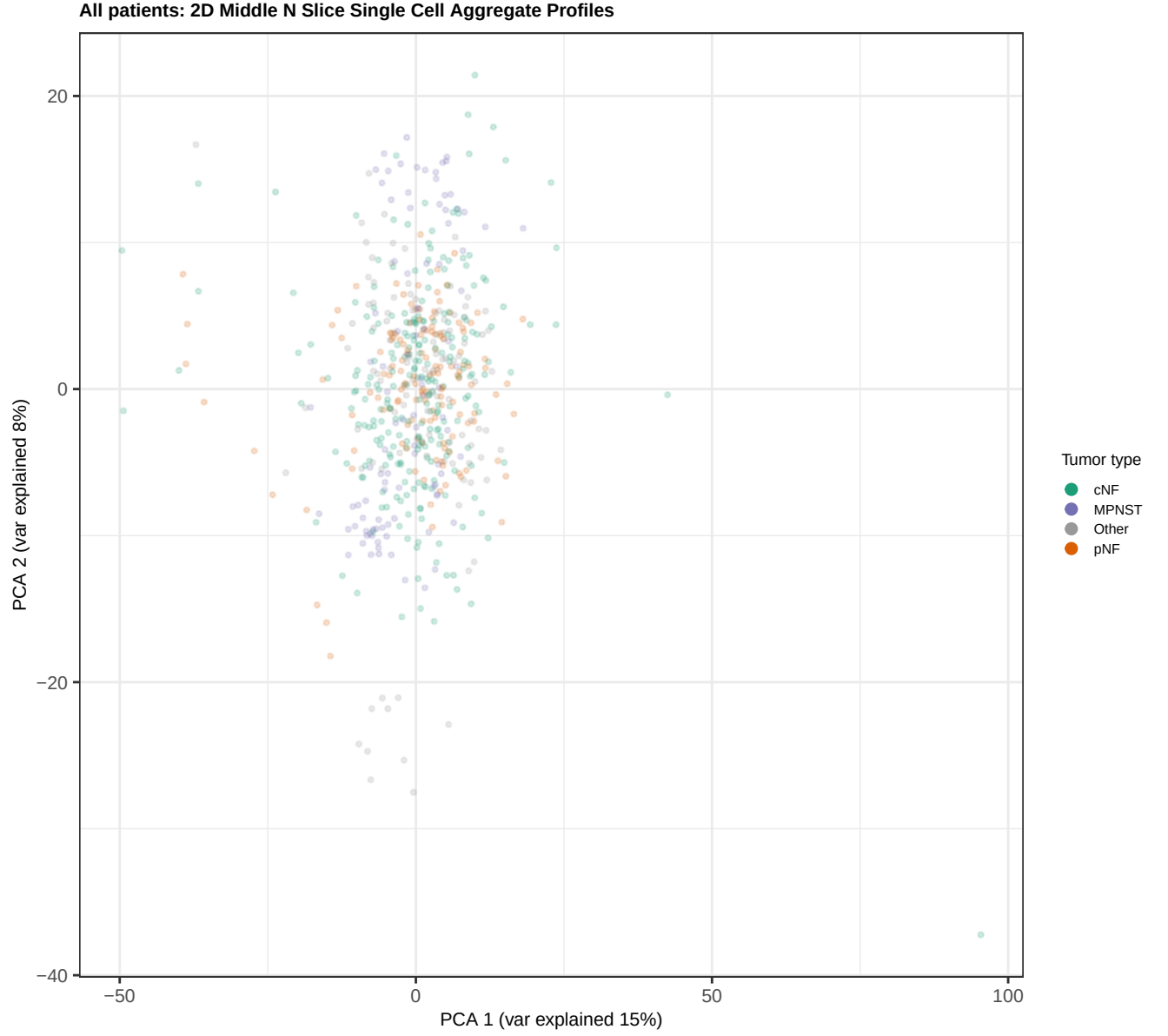

All patients: 2D Middle N Slice Single Cell FS Profiles
PCA 2 (var explained 6%)
50
0
−50
−50
0
50
PCA 1 (var explained 7%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

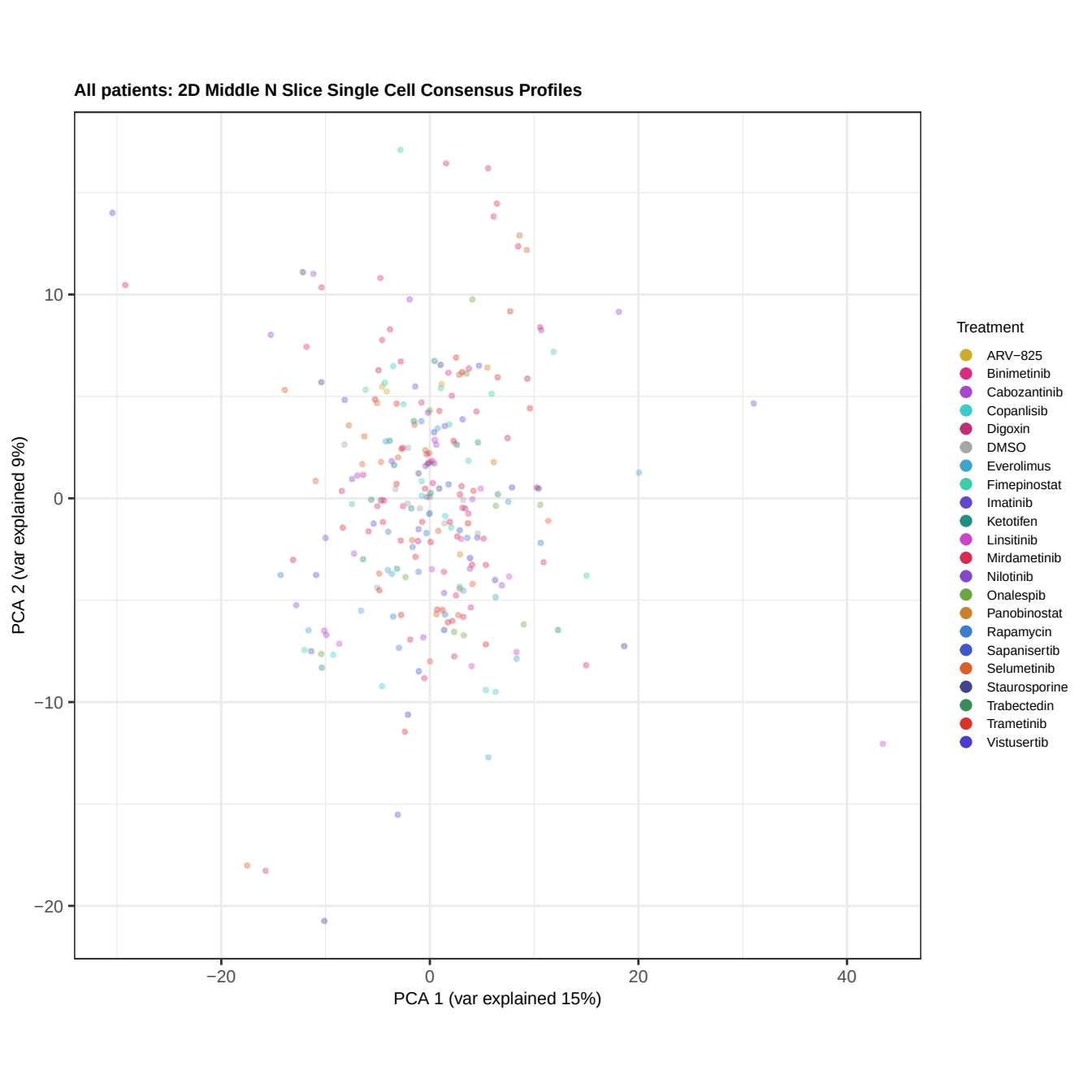

All patients: 2D Middle N Slice Single Cell FS Profiles
PCA 2 (var explained 6%)
50
0
−50
−50
0
50
PCA 1 (var explained 7%)
Tumor type
cNF
MPNST
Other
pNF

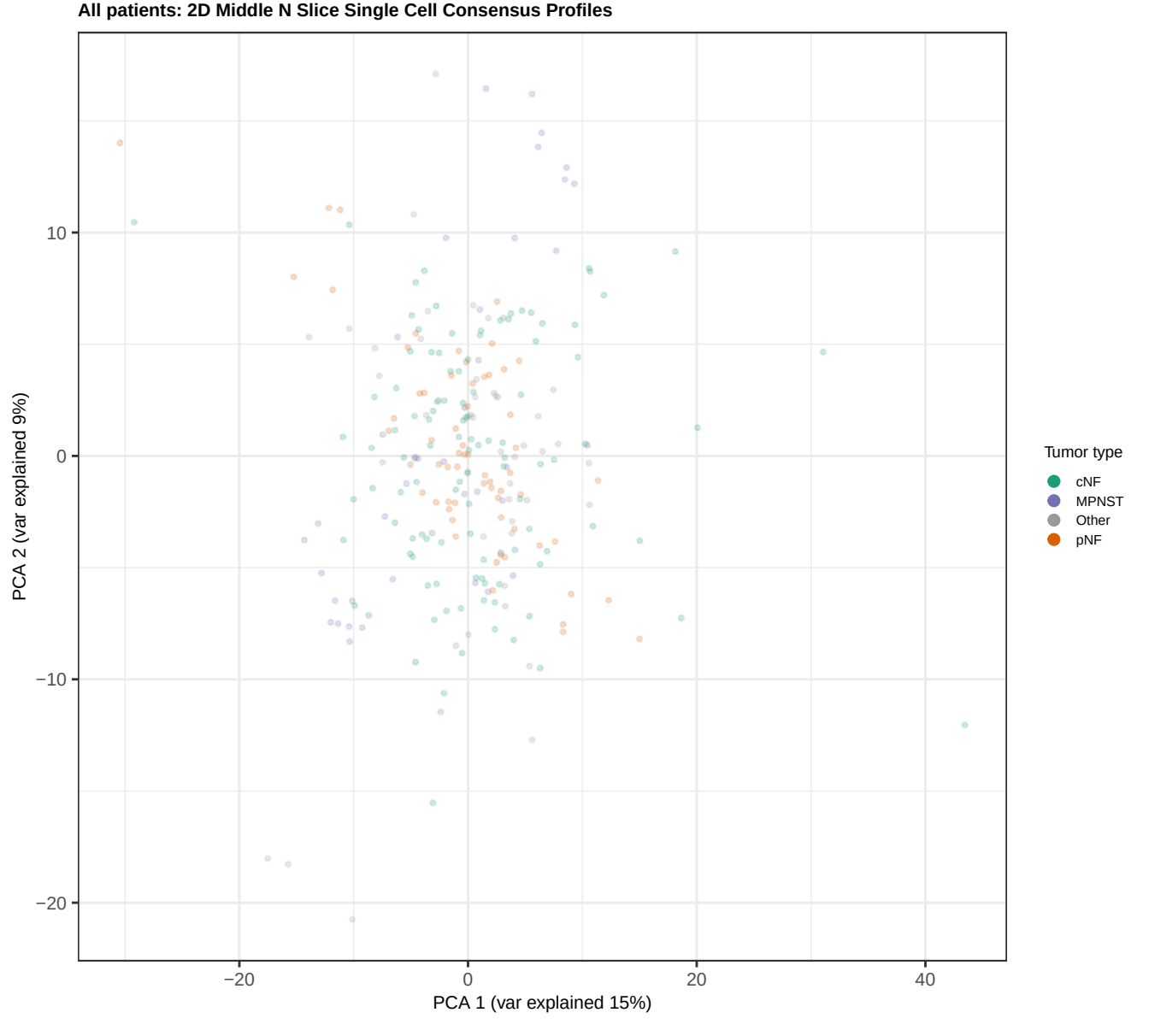

All patients: 2D Middle N Slice Single Cell FS Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
PCA 2 (var explained 6%)
PCA 1 (var explained 7%)
50
0
−50
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

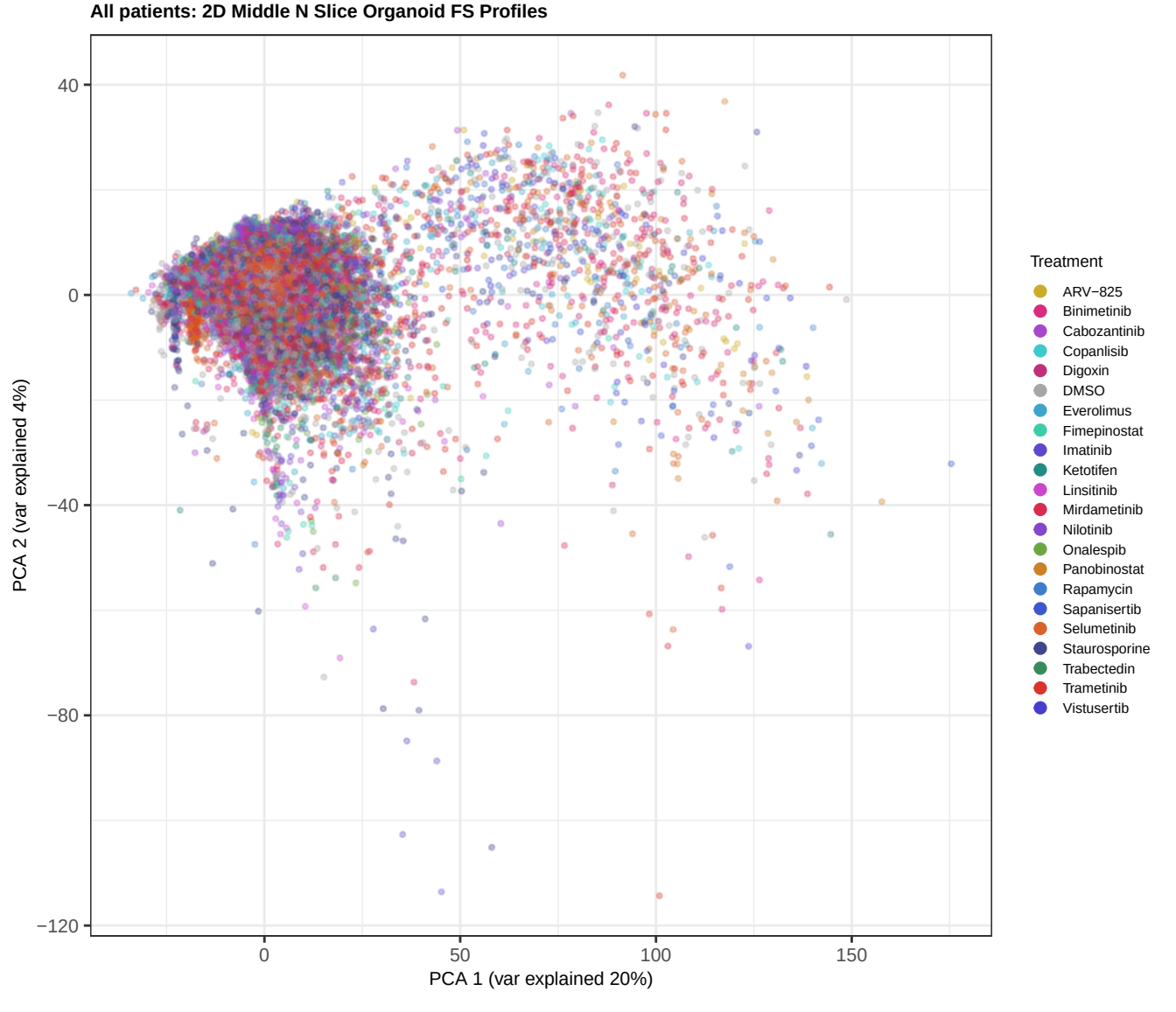

All patients: 2D Middle N Slice Single Cell FS Profiles
cNF
MPNST
Other
pNF
PCA 2 (var explained 6%)
PCA 1 (var explained 7%)
50
0
−50
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

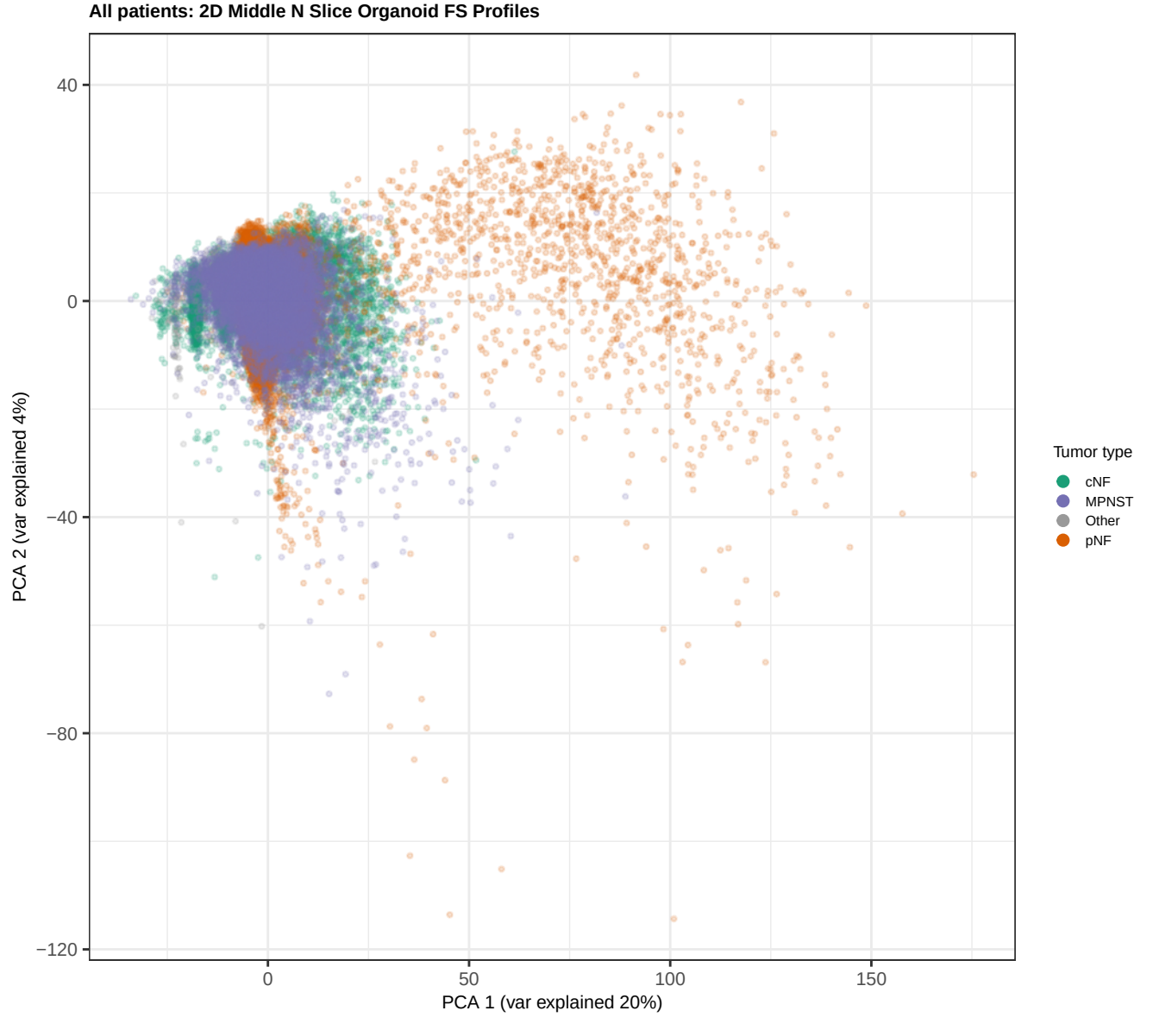

All patients: 2D Middle N Slice Single Cell Aggregate Profiles
20
0
−20
−40
PCA 2 (var explained 8%)
−50
0
50
100
PCA 1 (var explained 15%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

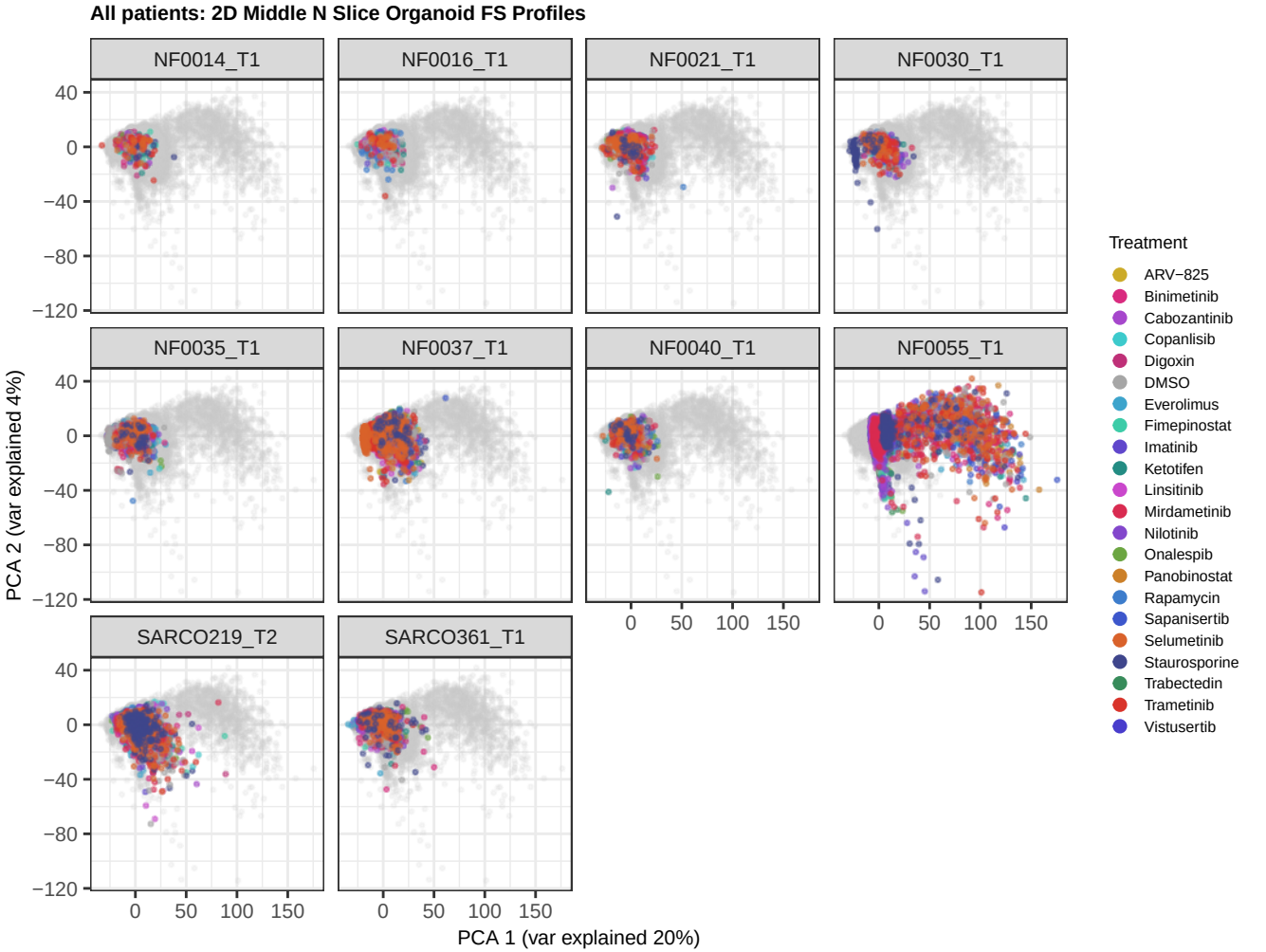

All patients: 2D Middle N Slice Single Cell Aggregate Profiles
20
0
−20
−40
PCA 2 (var explained 8%)
−50
0
50
100
PCA 1 (var explained 15%)
Tumor type
cNF
MPNST
Other
pNF

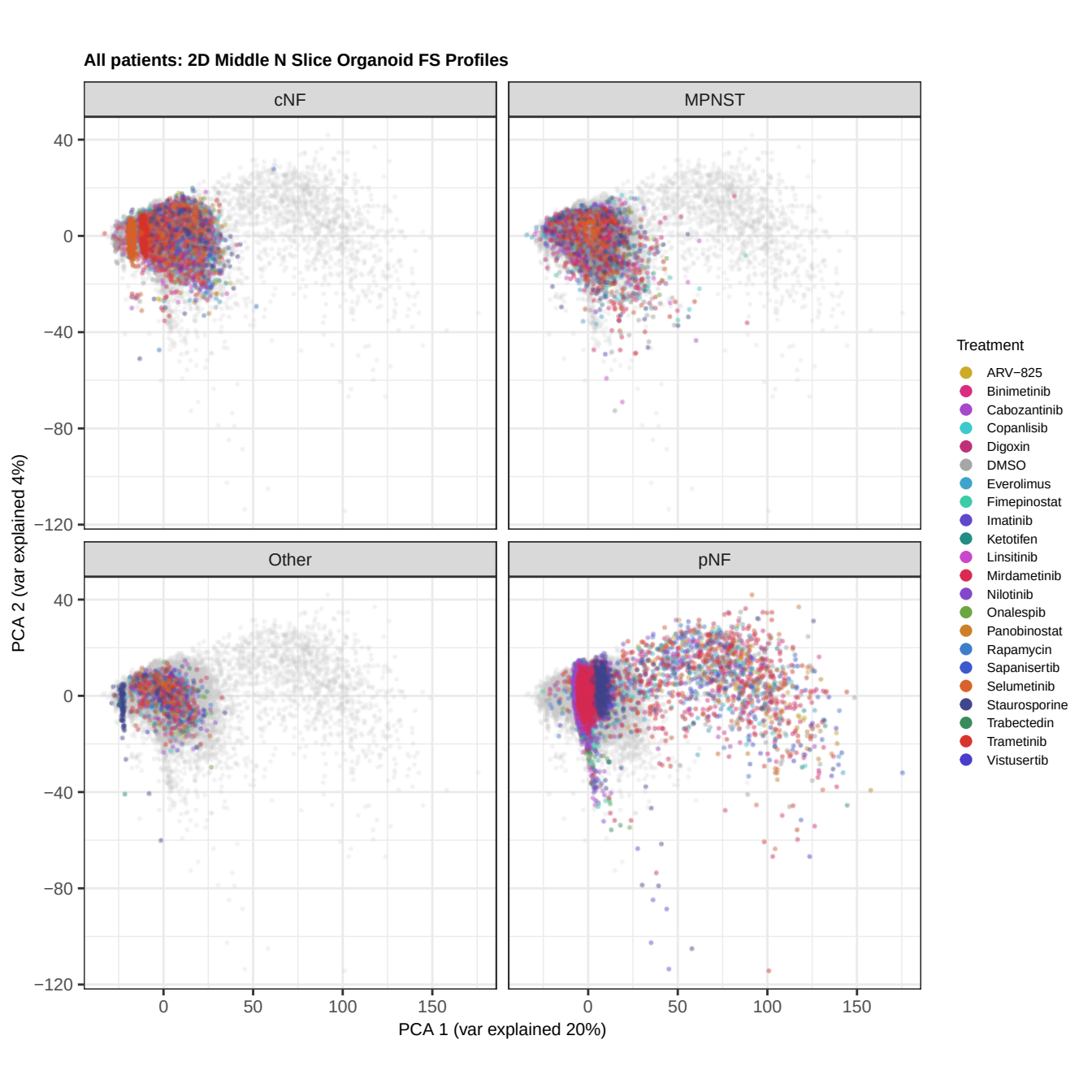

All patients: 2D Middle N Slice Single Cell Consensus Profiles
PCA 2 (var explained 9%)
10
0
−10
−20
−20
0
20
40
PCA 1 (var explained 15%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

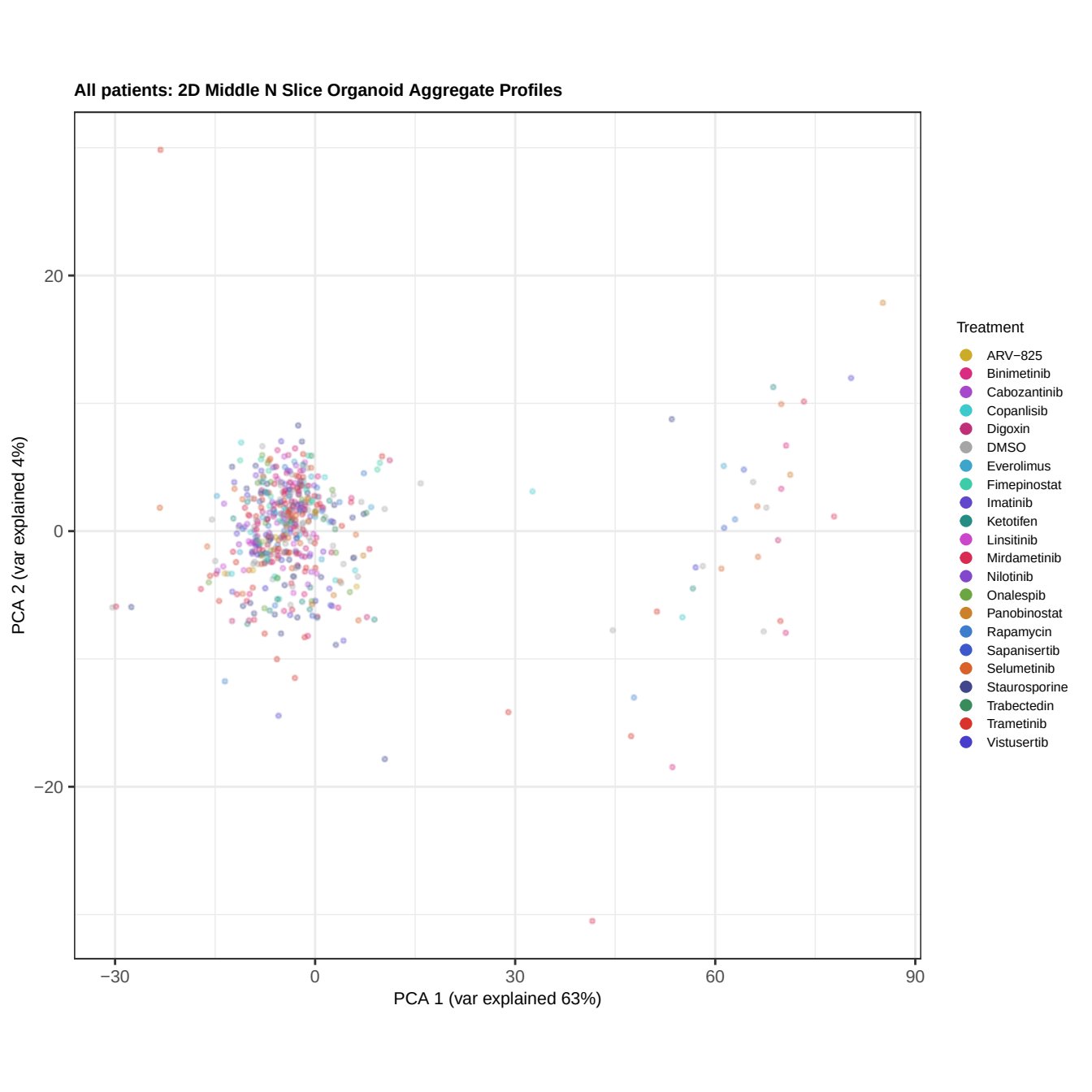

All patients: 2D Middle N Slice Single Cell Consensus Profiles
PCA 2 (var explained 9%)
10
0
−10
−20
−20
0
20
40
PCA 1 (var explained 15%)
Tumor type
cNF
MPNST
Other
pNF

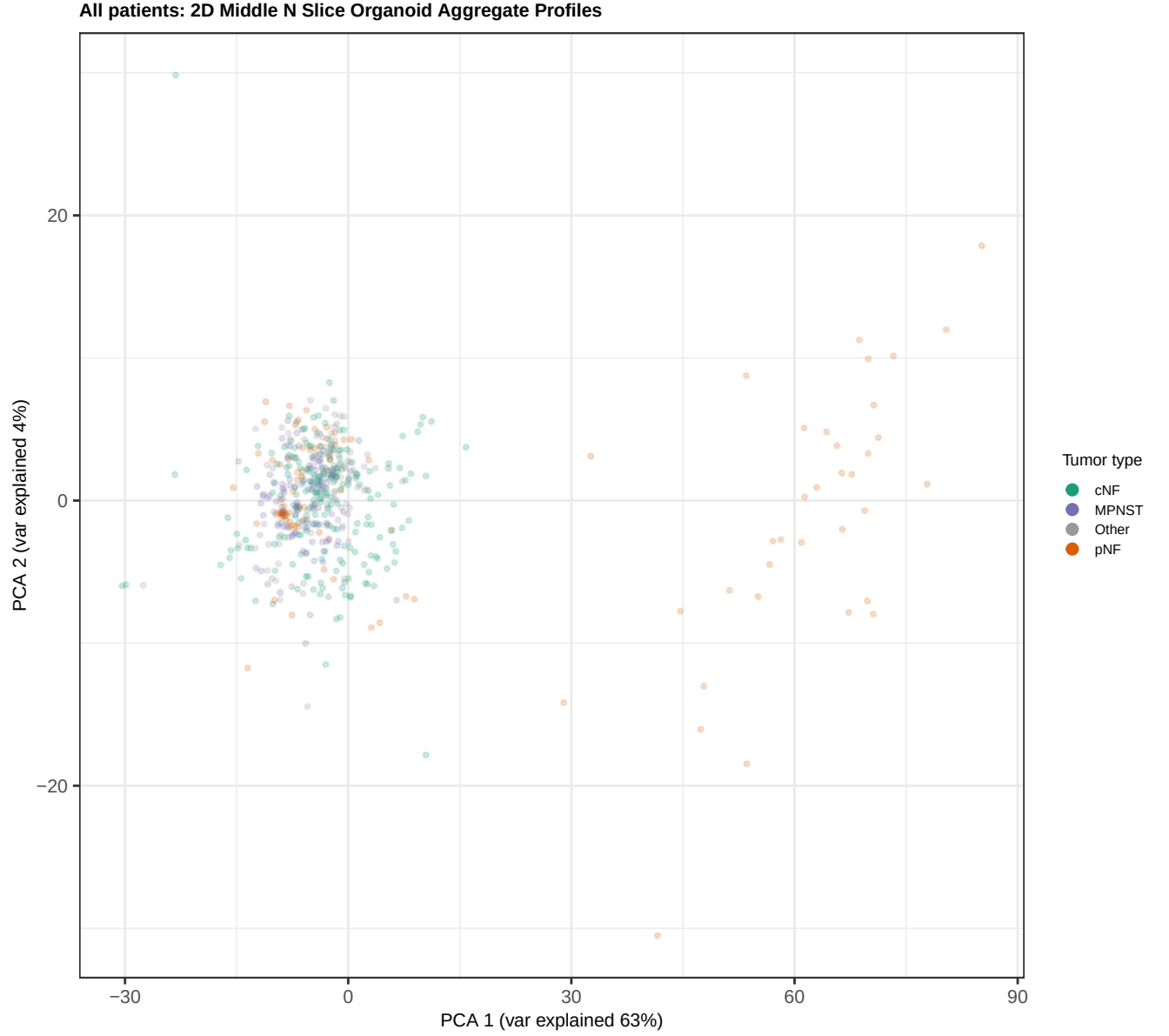

All patients: 2D Middle N Slice Organoid FS Profiles
PCA 2 (var explained 4%)
PCA 1 (var explained 20%)
40
0
−40
−80
−120
0
50
100
150
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

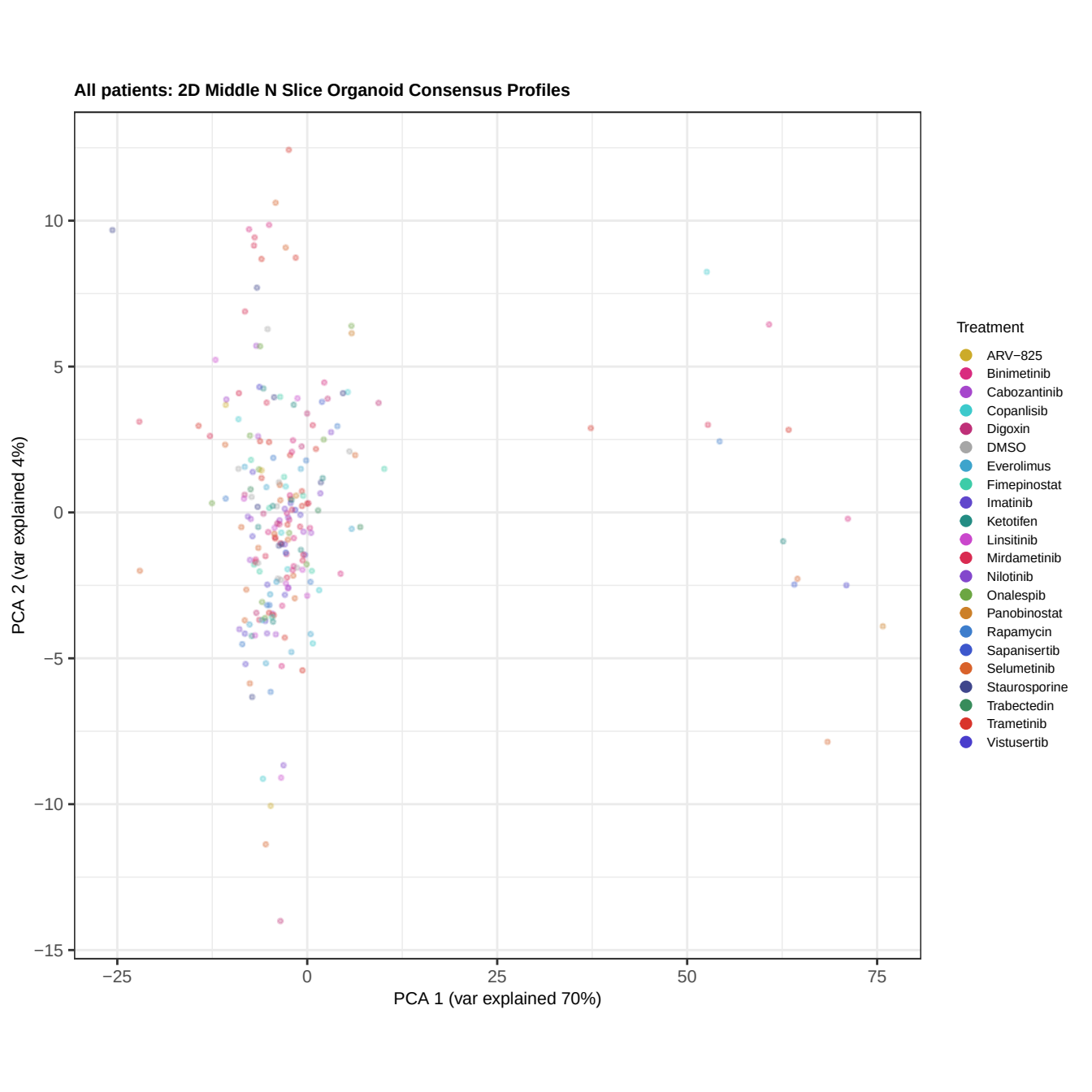

All patients: 2D Middle N Slice Organoid FS Profiles
40
0
−40
−80
−120
PCA 2 (var explained 4%)
0
50
100
150
PCA 1 (var explained 20%)
Tumor type
cNF
MPNST
Other
pNF

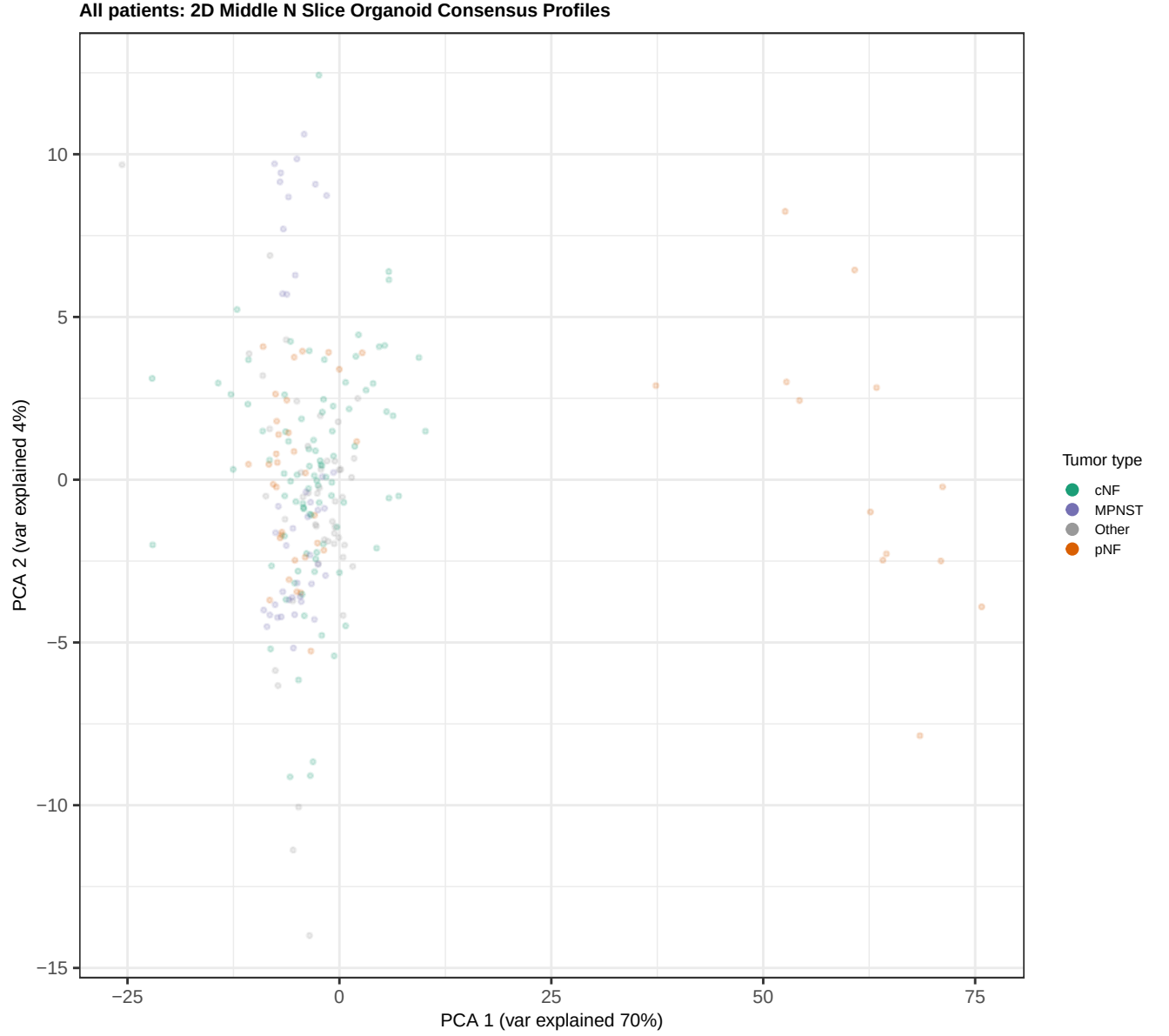

All patients: 2D Middle N Slice Organoid FS Profiles
NF0014_T1
NF0016_T1
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
PCA 2 (var explained 4%)
PCA 1 (var explained 20%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib


All patients: 2D Middle N Slice Organoid FS Profiles
cNF
MPNST
Other
pNF
PCA 2 (var explained 4%)
PCA 1 (var explained 20%)
40
0
−40
−80
−120
0
50
100
150
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib


All patients: 2D Middle N Slice Organoid Aggregate Profiles
PCA 2 (var explained 4%)
20
0
−20
−30
0
30
60
90
PCA 1 (var explained 63%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib


All patients: 2D Middle N Slice Organoid Aggregate Profiles
PCA 2 (var explained 4%)
20
0
−20
−30
0
30
60
90
PCA 1 (var explained 63%)
Tumor type
cNF
MPNST
Other
pNF


All patients: 2D Middle N Slice Organoid Consensus Profiles
PCA 2 (var explained 4%)
PCA 1 (var explained 70%)
10
5
0
−5
−10
−15
−25
0
25
50
75
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib


All patients: 2D Middle N Slice Organoid Consensus Profiles
PCA 2 (var explained 4%)
10
5
0
−5
−10
−15
−25
0
25
50
75
PCA 1 (var explained 70%)
Tumor type
cNF
MPNST
Other
pNF